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INTERNSHIP REPORT

**THEME: CONCEPTION AND REALIZATION OF AN INTEGRATED
NATIONAL BASED LEARNING APPLICATION FOR CAMEROON**

Internship carried out from the 03rd of July till the 30th of September 2025, in view of
obtaining a Higher Technician Diploma in Computer Sciences. Option: Software

Engineering

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Academic Year 2024 - 2025



CONCEPTION AND REALISATION OF AN INTEGRATED
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DEDICATION

THIS DOCUMENT IS DEDICATED TO MY
BELOVED MOTHER, MRS. MBO JOSEPHINE.



ACKNOWLEDGMENTS

Drafting this document would have not been possible without the contribution of some people who took upon themselves to see this work being accomplished. Our gratitude goes to the following people:

- **Mr. Armand Claude ABANDA**, the Resident Representative of the African Institute of Computer Sciences, for his unwavering support and guidance.
- **Mr. NDABOSE Daniel** chief executive officer of REALIZE company who accepted to give us the possibility to carry out our internship in his start-up.
- Our professional supervisor **Mr. MENGOT Nso** for his guidance and the necessary professional assistance he gave at every stage of the production of this report and the realization of this project during the internship.
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- All our teachers at the African Institute of Computer Sciences for the knowledge, they affected me with throughout the year making me to reach where I am today.
- The amazing family at REALIZE for their endless love and support during this internship period.
- My family most especially my mother for her guidance and support without which we could not attain the end of this work.
- All our classmates and friends at AICS for their knowledge, ideas, collaboration and most importantly the unforgettable moments spent together.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



ABSTRACT

In the Cameroonian context, particularly within the Anglophone regions, the education sector faces persistent challenges such as limited access to quality resources, socio-political disruptions, inadequate infrastructure, and a widening digital divide. To address these issues, this project was developed around the theme “**Conception and Realization of an Integrated National Based Learning Application,**” resulting in the creation of CamEdu, an innovative digital learning platform designed to ensure continuous, inclusive, and accessible education. Following a structured methodology based on feasibility studies, UML system analysis, and the 2TUP process, the application was developed using MongoDB, Express.js, React Native, and Node.js, resulting in a secure, scalable, and user-friendly solution. CamEdu demonstrates the potential of technology, digitalization, and innovation to improve access to education, support learning continuity, and strengthen resilience in crisis-affected regions.

Keywords:

- CamEdu
- Digital Learning
- Educational Technology



RESUME

Dans le contexte camerounais, particulièrement dans les régions anglophones, le secteur de l'éducation fait face à de nombreux défis tels que l'accès limité aux ressources de qualité, les perturbations socio-politiques, l'insuffisance des infrastructures et l'élargissement de la fracture numérique. Pour répondre à ces problématiques, ce projet s'est articulé autour du thème « **Conception et Réalisation d'une Application Nationale Intégrée d'Apprentissage** », donnant naissance à CamEdu, une plateforme numérique innovante conçue pour garantir un apprentissage continu, inclusif et accessible. En suivant une approche méthodologique structurée basée sur les études de faisabilité, l'analyse UML et le processus 2TUP, l'application a été développée avec MongoDB, Express.js, React Native et Node.js, offrant une solution sécurisée, évolutive et conviviale. CamEdu illustre le potentiel de la technologie, de la numérisation et de l'innovation pour améliorer l'accès à l'éducation, assurer la continuité pédagogique et renforcer la résilience dans les zones en crise..

Mots-clés :

- CamEdu,
- Apprentissage Numérique,
- Technologie Éducative



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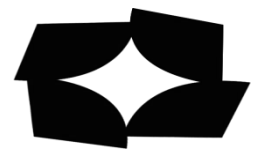
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ABBREVIATIONS

- **2TUP:** Two Track Unified Process
- **AICS:** African Institute of Computer Sciences
- **API:** Application Programming Interface
- **CSS:** Cascading Style Sheet
- **DBMS:** Database Management System
- **DIY:** Do It Yourself
- **GUI:** Graphical User Interface
- **HTML:** Hypertext Mark-up Language
- **HTTP:** Hypertext Transfer Protocol
- **IDE:** Integrated Development environment
- **JSON:** JavaScript Object Notation
- **KYC:** Know Your Customer
- **OS:** Operating System
- **UML:** Unified Modelling Language



GENERAL INTRODUCTION

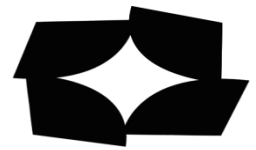
Despite the numerous challenges it has encountered, the field of computer science in Cameroon has steadily advanced over time. In today's context, individuals across the country continue to face significant difficulties in accessing effective educational services. Addressing these challenges has become crucial, as education remains a central pillar for personal development and national growth. In particular, Cameroon's Anglophone secondary education system has been deeply affected by resource disparities, teacher shortages, infrastructural gaps, and recurring disruptions in the continuity of learning. These persistent issues have discouraged many learners from fully engaging in basic education, thereby widening the educational divide. In this light, it becomes imperative to design innovative solutions that respond to these pressing challenges and help bridge the gaps in access, delivery, and quality of education. To contribute to this effort, the **African Institute of Computer Sciences (AICS)**, a pioneering institution in the field of technology, has established a three month internship program for Level 2 students. The program is designed to enable students to apply the knowledge acquired in the classroom to professional contexts, strengthen their technical and analytical skills, and support communities in overcoming educational challenges while promoting sustainable growth. As part of this program, our internship was carried out at **REALIZE**, a company specialized in software development and committed to fostering sustainable technological solutions. During this internship, we worked on the project entitled: **“Conception and Realization of an Integrated National Based Learning Application.”**

This project was structured into eight main stages:

1. **Insertion phase** – introducing the host company and documenting the intern's integration into the organization.
2. **Existing system** – presenting the current system in place.
3. **Specification book** – outlining user needs, project requirements, and constraints.
4. **Analysis phase** – applying the chosen analysis method and drafting the initial UML models of the system.



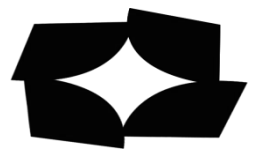
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5. **Conception phase** – studying the analyzed system in detail to identify real-world components.
6. **Realization phase** – implementing the system and making it available to users.
7. **Testing functionalities** – demonstrating the modules and features of the application.
8. **User guide** – providing instructions on system installation and usage.



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PART ONE: INSERTION PHASE



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PREAMBLE

The inserted document forms part of my internship report, which details the structure and objectives of the organization where I completed my internship. It presents an overview of my internship environment and delineates the different stages I underwent to successfully assimilate into the company.

CONTENT

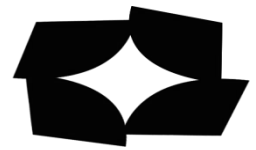
INTRODUCTION

- I. WELCOMING AND INTEGRATION
- II. GENERAL PRESENTATION OF REALIZE
- III. ORAGANIZATION OF REALIZE
- IV. HARDWARE AND SOFTWARE RESOURCES
- V. BREIF PRESENTATION OF PROJECT

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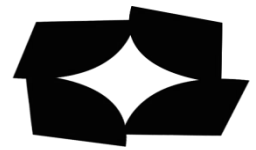


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INTRODUCTION

The insertion phase is a period (generally of 02 weeks) reserved for the different interns to discover and to familiarize with the working environment. Here, we got to know about the staffs, the different hardware and software resources used, the different departments which constitute the enterprise, how the company function both internally and externally and we were introduced to our workspace. During this period, we were also attributed an internship master often called professional supervisor and a theme. We also had a time to discuss among us interns on topics like what we love doing most, what we dislike, our beliefs and experiences. We shared about different realizations and failures in life.



I. WELCOMING AND INTEGRATION

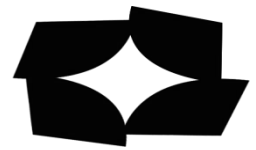
a. Welcome

The welcoming and integration event took place on the 3rd of July at 8:30 am. Upon arrival, Mr. Mengot, the senior developer, delivered an inspiring welcoming speech. Having interned at REALIZE this year, this moment felt familiar. We proceeded with a presentation and a meeting focused on discussing the challenges we faced during the coding test, which was a prerequisite for our internship at the company. The meeting also included an overview of the enterprise's structure, its internal policies, and the specific workstations assigned to each intern. Mr. Tanue concluded by encouraging us to strive for excellence and offered his support for any issues we might encounter.

b. Integration

After settling into our designated workstations, we were promptly assigned our first task for the day. During the first week, we conducted a verification of concepts using fundamental HTML and CSS to ensure everyone was on an even playing field before progressing. We then engaged in discussions about project ideas, where we were encouraged to propose our own.

In the second week, we focused on JavaScript concept verification exercises to assess our understanding and reasoning skills. We held a detailed discussion with our professional supervisor about the specification book and its various components, where he outlined what was expected of us in each segment. We also continued developing our project ideas into a concrete plan, benefiting from my previous familiarity with the company's expectations and processes.



II. GENERAL PRESENTATION OF REALIZE

i. Background

Realize is a Cameroon based non-governmental tech start-up institute founded by NDELOKAKEH Daniel in 2022, which proposes IT solutions and empowers the use of new technologies in Cameroon

ii. Missions, Visions and Activities of Realize

1. Missions:

The mission of Realize mainly relates to the empowerment and the perpetuation of Technology use. These missions include:

- Design and realize mobile-oriented software for companies and individuals;
- Assist new-born companies for quick growth using computer sciences;
- Provide training and certifications to improve qualified human resources in many fields of study;
- Take an active part in the sustainable development of the world through innovative solutions and virtual reality.

2. Vision:

At Realize, we believe that the true potential of every organization and individual can be unlocked through the harmonious fusion of analysis, development, realization, and data analytics. Our vision propels us forward, guiding us to make a lasting impact on the digital landscape and shape a future where innovation knows no bounds."

3. Activities:

The activities of Realize range from computer sciences, engineering, and training. We can outline the following:

- Conception realization and hosting of websites;
- Software development and maintenance;
- Training in Software related fields;
- Conception and realization of multimedia;
- IT consulting and innovation

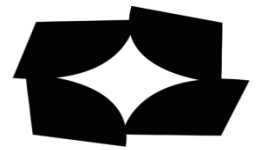


- IT support

4. Localization



Figure 1: Localization plan of Realize



III. ORGANIZATION OF REALIZE

A. Administrative organization

Realize if administratively organized as follows:

a. The General management

This is the highest level of the company, which takes care of the following:

- Proper functioning of each department;
- Define project strategies;
- Provide leadership and guidance to the company's employees.
- Makes critical decision that could affect the company's operations or reputation
- Severe as a point of contact for key stakeholders such as investors etc.

b. Human resource department

This department is in charge of the following:

- Recruitment and Hiring
- Acts as a liaison between employees and management;
- Manages employee compensation and benefits programs;
- Develops and implements company policies and procedures, ensuring compliance with employment laws and regulations.
- Manages administrative tasks related to employee records, data management, and HR systems

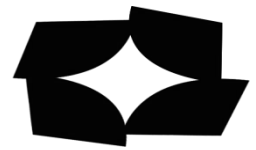
c. Communication Department

This department is in charge of the following:

- Handles public relations activities, which involve managing the company's reputation and image;
- Responsible for crafting and delivering messages to external stakeholders such as customers etc.;
- It fosters effective communication within the company;
- They create and manages content across different platforms and channels;



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d. Department of Financial Affairs

This department is in charge of the following:

- Responsible for developing and managing the company's financial plans and budgets;
- Prepare and presents accurate and timely financial reports to management, stakeholders, and regulatory authorities;
- Responsible for managing cost and expenses within the organization;
- Ensures compliance with tax laws and regulations. Etc.

e. Technical department

This department is in charge of the following:

- Responsible for managing the company's technological infrastructure, including networks, hardware;
- Play a role in managing and maintaining the company's data;
- Responsible for implementing and maintaining cybersecurity measures to protect the company's digital assets from potential threats;
- Interacts with technology vendors and manages relationships with external service providers. Etc.

f. Software Engineering Department

This department is in charge of the following:

- Primarily responsible for developing software applications and systems;
- Responsible for maintaining and supporting software application throughout their life-cycle;
- Evaluation and realization of projects etc.

B. FUNCTIONAL ORGANIZATION OF REALIZE

The functional branch of Realize is organized as follows

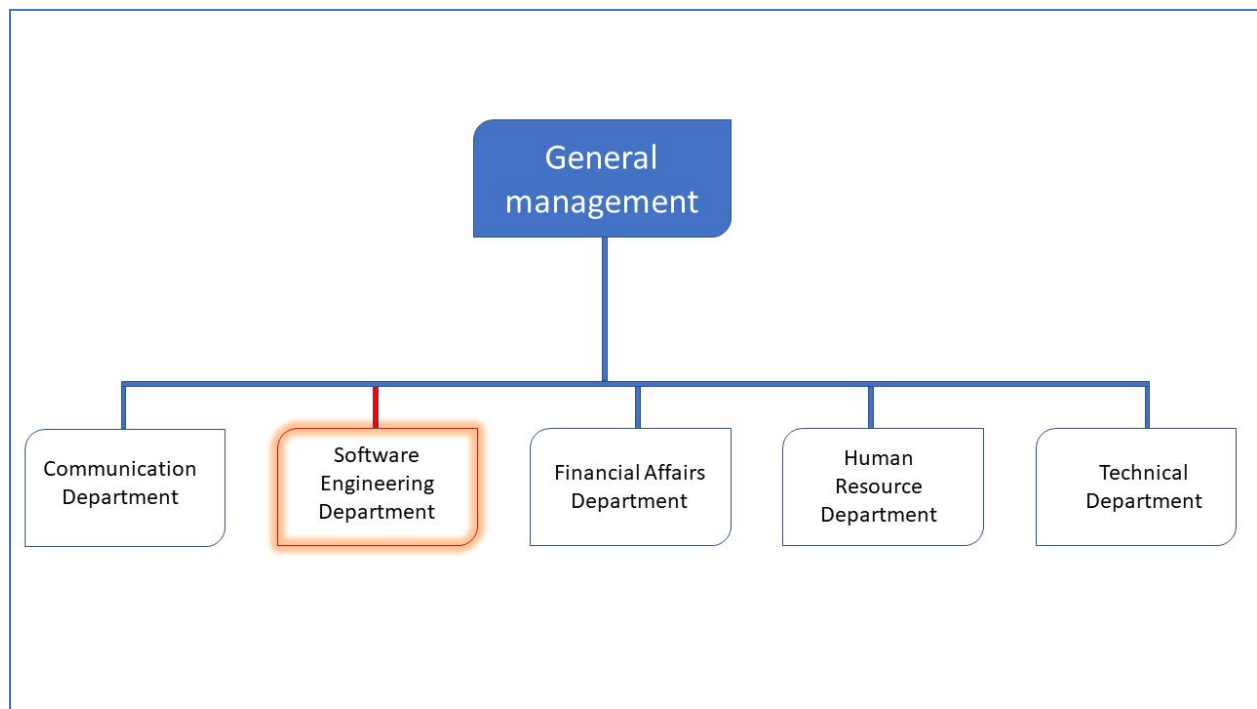
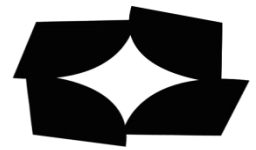


Figure 2: Functional organizational Chart of Realize



IV. HARDWARE AND SOFTWARE RESOURCES

A. HARDWARE RESOURCES

Table 1: Hardware resources

No	DESIGNATION	QUANTITY	CHARECTERISTICS	OBSERVATION
1	Lenovo	1	ThinkPad	Good
2	Modem	1	CAMTEL	Good
3	LCD Screen	1	HP	Excellent
4	Training equipment and office furniture	/	/	

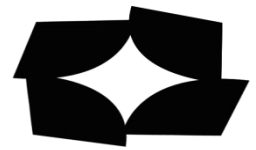
B. SOFTWARE RESOURCES

Table 2: Software resources

DESIGNATION	SOFTWARE
Operating System	Windows 11
Design tools	Photoshop
Integrated development environment (IDE)	Visual studio code
Text editor	Sublime text, notepad++
Database management system (DBMS)	MongoDB, PostgreSQL, MySQL
Web browser	Google chrome, Microsoft edge
Document editor	Microsoft office word



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON

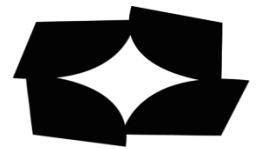


V. BRIEF PRESENTATION OF PROJECT

At the end of the initial two weeks (insertion period), we received a topic, namely **CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APP**. This project involves developing a platform designed to provide notes to students on several means like the reading form and the practical form, while providing questions to these forms to test the student. The primary goal of the application is to promote educational facility, increase literacy and simplify the access to a digitalized education world.



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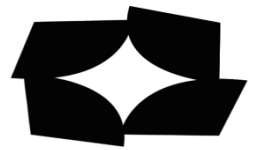


CONCLUSION

Our integration into the company was a pleasurable experience, with friendly and approachable colleagues and a professional supervisor who made us feel welcomed. Their availability and smooth integration into the company's operations were commendable. One of the most valuable takeaways from this phase was the importance of collaboration and discipline, crucial to any successful career. The subsequent phase of this momentous journey involves the project specifications, which must be precise to comprehensively understand what is needed to extract the most value from this transformative project.



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PART TWO: TECHNICAL PHASE



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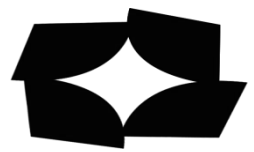


Content

DOCUMENT I: EXISTING SYSTEM
DOCUMENT II: SPECIFICATION BOOK
DOCUMENT III: ANALYSIS PHASE
DOCUMENT IV: CONCEPTION DOCUMENT
DOCUMENT V: REALIZATION PHASE
DOCUMENT IV: TEST OF FUNCTIONALITIES
DOCUMENT VI: USER GUIDE



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DOCUMENT I: EXISTING SYSTEM



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



PREAMBLE

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

CONTENT

INTRODUCTION

I THEME PRESENTATION

II STUDY OF THE EXISTING SYSTEM

III LIMITATION OF THE EXISTING SYSTEM

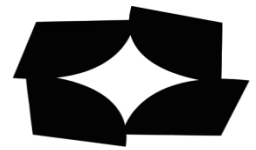
IV PROBLEMATIC

V PROPOSED SOLUTION

CONCLUSION



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INTRODUCTION

In this part of our report, we will explain our theme. We will also study the current system and identify its strengths and weaknesses. Based on this analysis, we will present the problem statement, which is the main challenge or gap that the current system faces. Finally, we will propose a solution that addresses this problem and improves the system.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



I. THEME PRESENTATION

The theme of this project is the Conception and Realization of an Integrated National Based Learning App designed to provide notes to students, offering local and reliable means of education based on the Cameroon program for the Anglophone Secondary Education. The app aims to simplify the way of learning of students, based on their specific needs. By offering both notes and past questions, the app caters to different people, from those seeking for their end of year examination, those from remote areas, people with limited resources and even, people with disabilities, especially those with reduced mobility.

Anyone in a need of acquiring more knowledge, be it for any national end of year examination, which are GCE Ordinary Level and GCE Advanced Level, be it for remedial classes, home classes or even in some professional career like teaching, the App is designed to provide the necessity means to each user to permit them attain their objectives. The platform do not only provides harmonized notes, but also exercises, past GCE questions for the past 10 years at least, solutions and answers, all that in a simplified designed to facilitate user experience and increase student literacy.



II. STUDY OF THE EXISTING SYSTEM

The development of a national learning platform for the Anglophone Secondary Education in Cameroon presents a transformative opportunity to address educational inequity, standardize quality for the education system. However, this ambition must be grounded in a realistic understanding of the existing infrastructure, socio-political challenges, and pedagogical needs.

In the current system, the educational landscape for the Anglophone Secondary Education is a patchwork of traditional methods nascent digital attempts. On the traditional classroom-based learning, the dominant model is face-to-face instruction in schools following the Cameroon GCE Board syllabus. Teaching relies heavily on chalkboards and textbooks.

Moreover, private online tutors where individual teachers uses WhatsApp, YouTube or Zoom to offer paid lessons, some educational websites like CameroonGCE.com, which offers past exams papers and notes which are often incomplete, unvented and not event integrated into a curriculum.

Furthermore, some government efforts, especially the Ministry of Secondary Education (MINSEC) has expressed intent and many have pilot programs, but there is no centralized functional platform for Anglophone schools. Some short-term projects being abandoned due to the often lack of sustainability.

Also, some e-learning platforms actually in activity are not local-based, most of them are created by foreigners and targets some specific courses, for instance Udemy and Khan Academy, which provides world-wide courses but not for the Cameroon program.



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III. LIMITATIONS OF THE EXISTING SYSTEM

Table 3: Limitations, Consequences and proposed solutions

LIMITATIONS	CONSEQUENCES	PROPOSED SOLUTION
Lack of Standardization	Existing online resources are unregulated, varying greatly in quality and accuracy which may not perfectly align with the official GCE syllabus.	Syllabus-Aligned content for digitalized official curriculum with text, videos, explanations and quizzes.
Teachers Preparedness	Many teachers lack digital literacy skills to create or use e-learning tools effectively and pedagogic training for online instruction is rare.	Mandatory Teacher Training to integrate digital pedagogy into teacher training colleges (ENS) and run it national-wide to permit teachers on how to use such platform effectively.
Internet Connectivity	Unreliable and expensive in many rural areas where data cost is prohibitive for most students and teachers.	Offline functionality in which the app is designed to allow users to download full courses and lessons when they have Wi-Fi or data access for or use later without connectivity.



IV. PROBLEMATIC

The study of the functioning of the existing system has permitted us to identify certain critics. As a result of those different critics, there major problem is **“How can we design and implement a national digital learning platform for Anglophone secondary schools in Cameroon?”**

V. PROPOSED SOLUTION

In order to solve the problems stated above, we are to realize an application based on the following points:

1. Develop a Centralized National Learning Platform

Create a mobile application dedicated to Anglophone secondary education, aligned with the official GCE syllabus. This platform will unify resources, improve standardization, and provide a reliable, long-term solution to fragmented learning.

2. Ensure Syllabus-Aligned Digital Content

Integrate curriculum-based content, including lessons, explanatory videos, notes, exercises, and quizzes, designed specifically to match the Cameroon GCE syllabus. This ensures both quality and uniformity in learning materials.

3. Teacher Capacity Building

Provide training modules and integrated tutorials to improve teachers' digital literacy and equip them with the skills to effectively use and contribute to the platform, fostering long-term sustainability.

4. Enable Offline Access and Lightweight Design

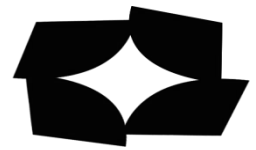
Incorporate offline functionality to allow students and teachers to download complete lessons or courses for later use without internet connectivity. The platform will be optimized to work smoothly on low-end devices with minimal data usage.

5. Improve Collaboration and Communication

Add features such as discussion forums, messaging, and feedback channels to encourage interaction between teachers, students, and administrators, while building a more engaging learning environment.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON

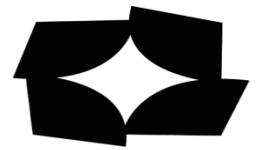


6. **Implement Role-Based Access and Data Security**

Adopt a secure authentication and authorization system where each user (student, teacher, and administrator) can only access the features relevant to their role, ensuring privacy, accountability, and trust.



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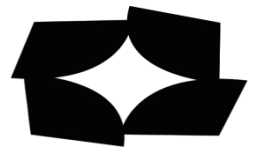


CONCLUSION

The theme is a vital component of the project as it helps to define the scope of work and identify the system under consideration. The study of the existing system is equally crucial. This is because, in order to improve a system, it is necessary to thoroughly examine it, identify its existing problems and limitations, and critically analyze its shortcomings. Once the study is completed and the problematic areas are identified, proposed solutions can be developed to address and resolve the issues that have been discovered.



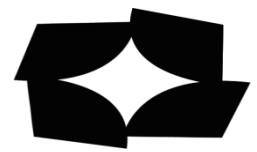
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DOCUMENT II: SPECIFICATION BOOK



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



PREAMBLE

The specification book serves as a contractual document before the application development process commences. It succinctly outlines the expectations of the project owner and the respective responsibilities of all parties involved. It provides a specific and detailed description of the topic and user requirements, as well as the necessary conditions for project execution. The project's success will depend on careful allocation and management of the various resources available, including physical, financial, and human assets. Therefore, this document outlines the criteria that the developer must adhere to in order to ensure the success of the project.

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INTRODUCTION

I CONTEXT AND JUSTIFICATION

II OBJECTIVES OF THE PROJECT

III EXPRESSION OF NEEDS

IV PROJECT PLANNING

V ESTIMATED COST OF THE PROJECT

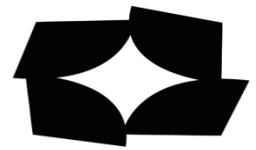
VI PROJECT CONSTRAINTS

VII LIST OF PARTICIPANTS AND DELIVERABLES

CONCLUSION



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



INTRODUCTION

The report's specification book enables us to provide comprehensive details about our topic, enhancing our understanding and improving its chances of success. To establish the project's scope, we will describe the context of our topic and identify the issues within it that we aim to solve. Once the solution is proposed, we will outline the objectives we have set for ourselves and for the project. In addition, we will assess the functional and non-functional requirements of our system before designing a plan to ensure timely completion of the project and determine the financial resources needed. Subsequently, we will outline the expected project deliverables at the conclusion of the project.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



I. CONTEXT AND JUSTIFICATION

1. CONTEXT:

Education is a fundamental pillar for national development, yet in Cameroon, particularly within the Anglophone regions, secondary education continues to face persistent challenges. These include limited access to standardized learning materials, frequent disruptions caused by socio-political crises, inadequate infrastructure, and poor digital integration. Traditional classroom-based instruction, heavily reliant on chalkboards and textbooks, remains dominant. However, this approach has proven insufficient in addressing the growing demand for flexible, inclusive, and resilient learning methods. While some private initiatives exist, such as online tutoring through WhatsApp, YouTube, Zoom, or platforms like *CameroonGCE.com*; they often lack curriculum alignment, quality assurance, and sustainability. Government efforts, though commendable, have largely been fragmented and unsustainable, with no centralized functional platform specifically designed for Anglophone secondary education. Meanwhile, global platforms such as *Udemy* and *Khan Academy* offer valuable resources but fail to contextualize content for the Cameroon GCE curriculum. In response, there is a pressing need for a **localized, syllabus-aligned, and technologically robust learning platform** that bridges the educational gap and promotes continuity of learning in times of crisis.

2. JUSTIFICATION

The justification for this project lies in the multiple deficiencies of the existing system, which include unstandardized resources, lack of teacher preparedness, and connectivity challenges. These issues have weakened the continuity and effectiveness of secondary education in Cameroon.

This project seeks to provide a sustainable solution through **CamEdu**, a national digital learning platform designed to:



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- **Standardize learning content:** Align resources with the official GCE syllabus, including lessons, videos, exercises, and quizzes.
- **Increase accessibility:** Provide both online and offline access, ensuring students in low-connectivity areas are not left behind.
- **Empower teachers:** Offer digital tools and resources to support teachers in lesson delivery and capacity building.
- **Foster trust and sustainability:** Ensure a secure, locally built, and culturally relevant platform tailored to Cameroon's educational context.

By leveraging modern technologies, the platform ensures a lightweight, secure, and scalable solution. This initiative will not only address current challenges but also pave the way for long-term digital transformation in Cameroon's secondary education system.

II. OBJECTIVES OF THE PROJECT

1. General Objective

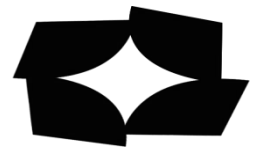
The overarching goal of this project is to design and implement a centralized national digital learning platform (CamEdu) that enhances accessibility, standardization, and continuity of quality education for Anglophone secondary schools in Cameroon.

2. Specific Objectives

- ✓ **Facilitate Accessible Learning:** Provide students with syllabus-aligned lessons and quizzes, available both online and offline.
- ✓ **Enhance User Experience:** Design an intuitive, mobile-friendly platform to meet the needs of learners and teachers.
- ✓ **Ensure Standardization and Transparency:** Align all digital resources with the official GCE curriculum.
- ✓ **Empower Teachers:** Equip educators with tools for digital teaching, lesson planning, and student engagement.
- ✓ **Enable Progress Tracking:** Integrate dashboards to monitor student progress and learning outcomes.



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- ✓ **Promote Collaboration:** Incorporate communication features (forums, messaging) to encourage student-teacher interaction.
- ✓ **Guarantee Security:** Implement role-based access to protect user data and maintain system integrity.

III. EXPRESSION OF NEEDS

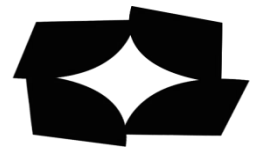
1. FUNCTIONAL NEEDS

In this context, functional requirements describe what the system or application should do. The modules are as follows;

1. Visitor
 - Create an account
 - Browse course list
2. Student
 - Register for a course
 - View course
 - View lecture
 - View course progress
 - Take exercise
 - View exercise solution
 - Make payment
 - Make payment with **OM (Orange Money)**
 - Make payment with **MOMO (Mobile Money)**
 - Authenticate (for secure access)
 - Access premium functionality
 - View past questions
 - View past questions & answers
 - View live practical



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3. Teacher

- Manage course material
 - Add course material
 - Update course material
 - Delete course material
- Manage exercises
 - Add exercise
 - Update exercise
 - Delete course exercise
- Make suggestion

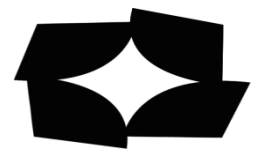
4. Admin

- Manage course
 - Add course
 - Update course
- Manage user account
 - Add user account
 - Delete user account

2. NON-FUNCTIONAL NEEDS

This is analyzing the needs our application is going to answer to. Our application must permit the DAG and the personnel, using the new system to have the following.

- **Reliability:** Ensure consistent and dependable performance with minimal downtime.
- **Portability:** Be accessible across various devices and platforms.
- **Speed:** Provide quick response times for user actions and data processing.
- **Security:** Robust authentication, data privacy compliance.
- **User-Friendliness:** Offer an intuitive and easy-to-navigate interface.
- **Flexibility:** Adapt to different user needs and scales with organizational growth



IV. ESTIMATED COST OF THE PROJECT

a. Software Resources

Table 4: Software resources (source: MERCURIALE-2025)

RESOURCES	DESIGNATION	USAGE	QUANTITY	UNIT COST (FCFA)
Formatting	MICROSOFT OFFICE PROFESSIONAL 2019	Writing of the report	1	355465
Web browser	Google Chrome	Access and view web pages on the internet	1	Freeware
Code Editor	Visual Studio Code	For writing the code of the application	1	Freemium
Project Planning	Gantt Project	For building a Gantt chart	1	Freemium
Graphic Editor	Icongram	For building a geographical location of the enterprise	1	Freemium
UML Analysis	Visual Paradigm	For drawing UML diagrams	1	Community Version
Version Control	Git & GitHub	For tracking and managing changes to software code	1	Free Version
TOTAL 1			7	355465



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b. Hardware Resources

Table 5: Hardware resources (source: MERCURIALE-2025)

RESOURCES	HARDWARE	QUANTITY	UNIT COST (FCFA)
Computer	DELL LATITUDE 7490 CORE i7 vPro 8 th Gen, HARD DISK 500 SSD, RAM 16GB	1	1295000
Printer	Printers	1	514528
Network	Local network installation	1	300 000
Removable Disk	USB HP 2.2 FLASH DRIVE V225W - 32 GB	1	14160
Smartphone	Sony Xperia X2Z Compact	2	150 000
Network	Local network installation	1	300 000
Total 2		7	2,573,688

c. Human Resources

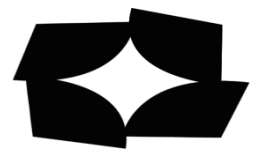
Table 6: Human resources (source: Mercurial 2025)

ROLE	NUMBER OF	QUANTITY	COST PER	TOTAL PRICE
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Written by WOUANSI WANSI SIMON PIERRE, student in level II at the African Institute of Computer Sciences (AICS) - Cameroon Paul Biya Technological Center of Excellence Academic year 2024-2025



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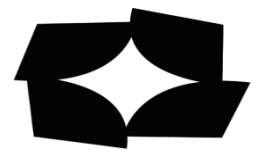


	DAYS		DAY	(FCFA)
Project Manager	90	01	30 000	2 700 000
Analyst	21	01	25 000	525 000
UI/UX Designer	30	01	10 000	140 000
Programmer	30	01	15 000	450 000
Tester	14	02	10 000	140 000
Margin error	/	/	/	1,475, 000
TOTAL 3				4 840 000

d. Global Estimation

Table 7: Global estimation

TOTAL 1 (FCFA)	TOTAL 2 (FCFA)	TOTAL 3 (FCFA)	OVERALL TOTAL (FCFA)
355,465	2,573,688	4,840,000	7,769,153
SEVEN MILLION SEVEN HUNDRED SIXTY-NINE THOUSAND ONE HUNDRED FIFTY-THREE			



V. PROJECT PLANNING

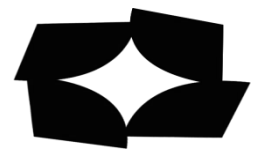
a. Chronogram of activities

Table 8: Chronogram of activities

PHASE	OBJECTIVE	OUTPUT	DURATION	PERIOD
INSERTION	Collection of information on the enterprise	Insertion book	2 weeks	03 rd July to 05 ^c th July
EXISTING SYSTEM	Study of the existing system	Existing System	1 week	22 nd July to 26 th July
SPECIFICATION BOOK	Specification of the user needs	Specification Book	1 week	29 th July to 02 nd August
ANALYSIS	Capture of needs Use case and textual description Modelling	Analysis Book	2 weeks	05 th August to 16 th August
CONCEPTION	Preliminary conception and Detailed conception	Conception book	2 weeks	19 th August to 30 th August
REALIZATION	Implementation Unitary test Integration Test Development	Realization book	3 weeks	02 nd September to 20 th September
TEST OF FUNCTIONALITIES	Testing of the software and debugging	Test of functionalities	1 week	21 st September to 27 th September
INSTALLATION AND USER GUIDE	Documenting software	User Guide	1 week	28 th September to 30 th September



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VI. LIST OF PARTICIPANTS AND DELIVRABLES

1. LIST OF PARTICIPANTS

Table 9: List of participants

NAME	FUNCTION	ROLE
Mrs. NGOUETSO Sandra	Follows up interns at the company level	Professional Supervisor
Mrs. NZALI Claudia	Follows up student at the academic level	Academic Supervisor
WOUANSI WANSI SIMON PIERRE	AICS Student Intern at Realize	Student at AICS



a. Gantt Project Planning

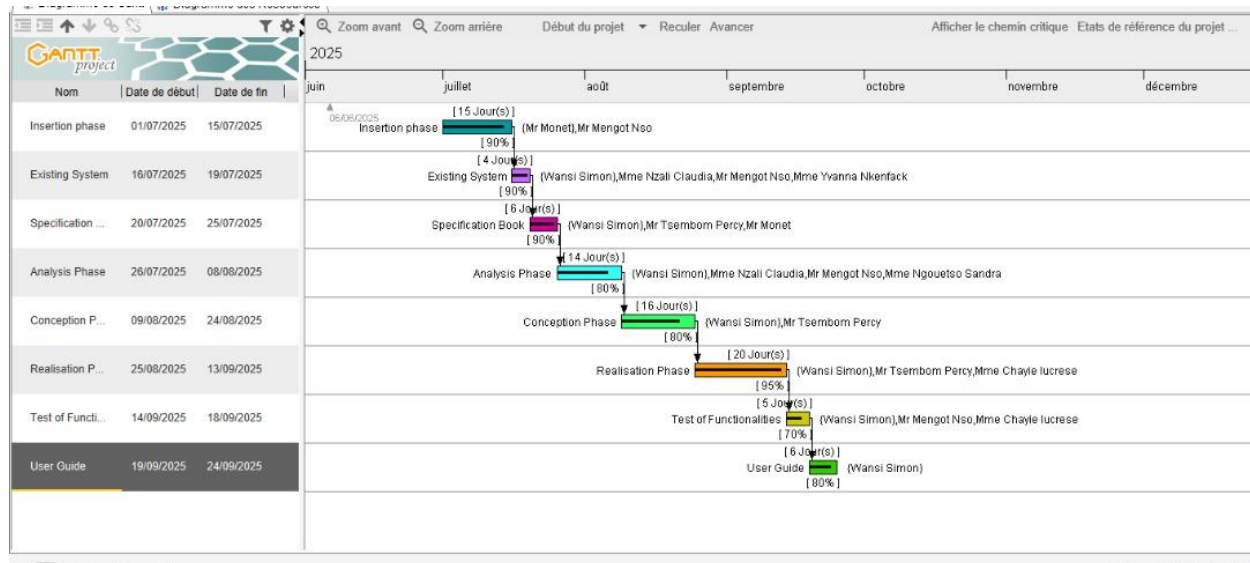
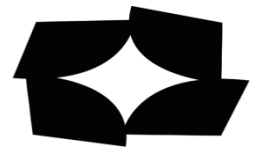


Figure 1: Project Gantt chart

2. DELIVERABLES

In project management, any component materializing the result of a realization service is called a deliverable. In the case of our project, the deliverables are: A report composed of the following document

- ❖ A User Guide
- ❖ Software Application
- ❖ PowerPoint

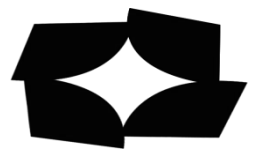


CONCLUSION

In conclusion, this part has come to a close, wherein we have effectively listed and expounded on our goals. The specifications book provided us with the means to present the project's different stakeholders, as well as the prerequisites and tentative schedule essential for its realization. Moving forward, we will directly proceed with the analysis phase, where we will utilise a modelling language and a unified approach to exemplify our system. Furthermore, we will carry out a comparative evaluation of UML and Merise methodologies.



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DOCUMENT III: ANALYSIS PHASE



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PREAMBLE

The primary aim of the analysis phase is to comprehend the user's requirements, define the scope of study and gain a thorough understanding of the system under examination. To accomplish this goal, we will employ the Unified Modelling Language (UML) with the 2 Track Unified Process (2TUP) methodology to analyse the system. Our approach will involve a comparison between UML and MERISE and various other unified processes. Lastly, we will introduce a solution model for our proposed software.

CONTENT

INTRODUCTION

I COMPARATIVE STUDY OF UML AND MERISE

II CHOICE OF THE ANALYSIS METHOD

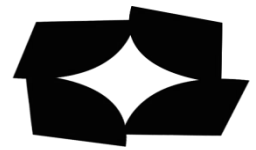
III JUSTIFICATION OF THE ANALYSIS METHOD

III MODELLING OF THE PROPOSED SOLUTION IV

CONCLUSION

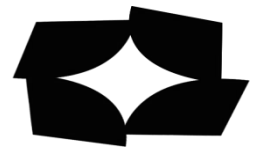


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INTRODUCTION

System development can be thought of as having two major components: System analysis and system design which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems has most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE unified processes and finally the various diagrams that meet the functional need requirements.



I. COMPARATIVE STUDY OF UML AND MERISE

a. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

b. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real world objects as a first step in developing an object-oriented design methodology. Its notation is derived from and unifies the notations of three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.0 comprises of 13 diagrams which represent the different views of a system. The 13 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include;

1. STATIC OR STRUCTURAL DIAGRAMS

-  Class diagram
-  Object diagram



Component diagram

Deployment diagram

Package diagram

Profile diagram

2. BEHAVIORAL PR DYNAMIC DIAGRAMS

Use case diagram

Activity diagram

State machine diagram

Sequence diagram

Communication diagram

Global Interaction diagram

Timing diagram

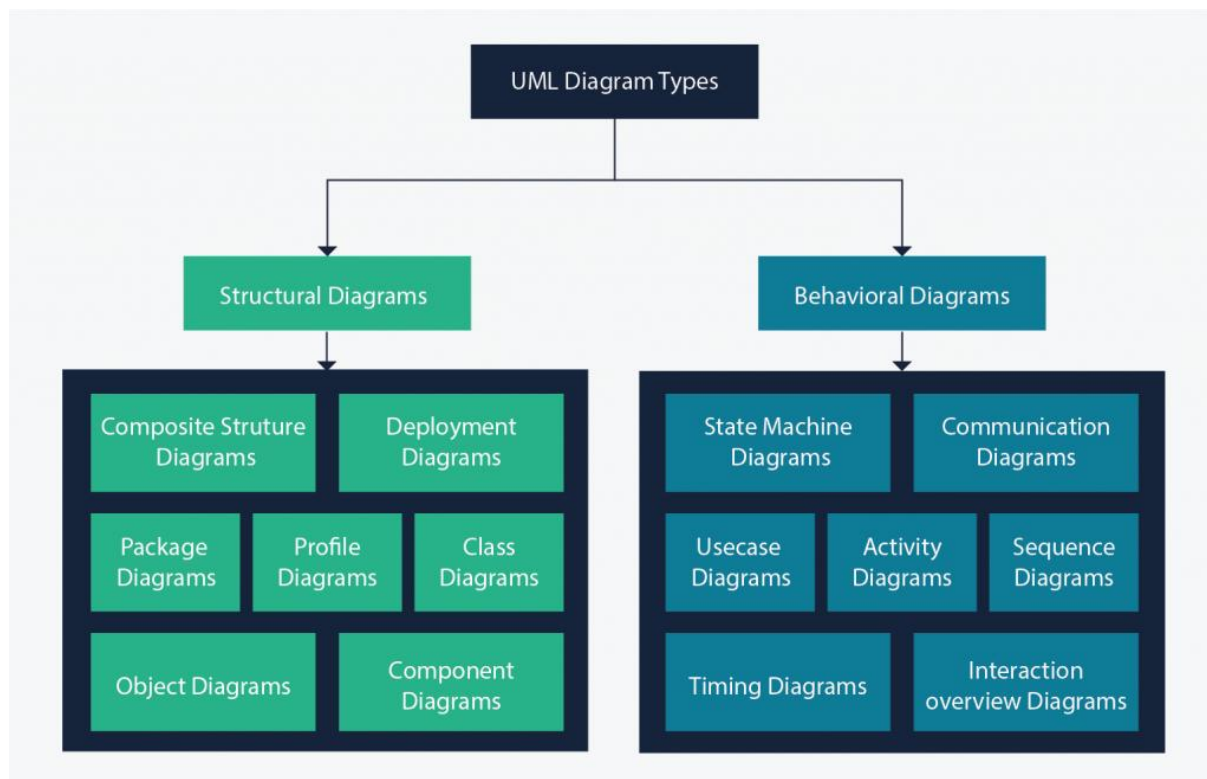
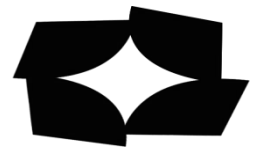


Figure 2: UML 2.5 diagrams overview (source: <https://creately.com/blog/diagrams/uml-diagram-types-examples/>)



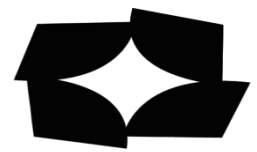
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On very important notice is that UML is not a method but a modelling language. As such to give it an approach we need to associate UML to a Unified Process (UP) in other to give our conception a methodology to follow. There exist several Unified Processes but our modelling approach will be the 2TUP (Two-Track unified process) which we will use in the course of our project.

Table 10: Comparison between MERISE and UML

MERISE	UML
MERISE is a systemic method of analysis and design of information systems. That is, it uses a system approach	UML is however not a method but rather an object modelling language to which it is necessary to associate an approach to make it a method. This is the case with the 2TUP method; RUT and XP.
Emphasizes data analysis and modelling using data flow diagrams (DFD), entity-relationship diagrams (ERD), and data dictionaries	Emphasizes system behaviour and structure using 14 diagram types, such as use case diagrams, class diagrams, and activity diagrams
Suitable for relational databases design and implementation	Suitable for object-oriented design and implementation



II. COMPARATIVE STUDY OF UNIFIED PROCESS

a. Unified Process

A Unified Process is a process of development of software constructed on UML; it is iterative, incremental, centered on architecture, driven by use cases and requirements.

Iteration is distinct sequence of activities with a basic plan and evaluation criterion that produces an internal or external output. Either the content of an iteration is improved, or the evolution of the system is evaluated by users.

An increment is the difference between two released products at the end of two iterations. Each iteration that the group is capable of integrating the technical environment in order to develop a final product and give users the possibility of having tangible results.

Centered on architecture the different models derived during the establishment of system must be reliable and coherent.

Driven by use case and requirements enables the clear definition of a users' needs and priorities respectively thereby minimizing the risk of project failure.

b. Two Track Unified Process (2TUP)

2TUP is a unified process which is built on UML and has as objective to bring solution to constraints of functional and technical changes imposed on information systems by strengthening controls on development capacities. It proposes a Y-shaped development life cycle that separates the functional aspect from the technical aspects, and the merging of these two forms the implementation aspect. 2TUP distinguishes therefore two branches: the functional and technical branches, the combination of the result of these two branches forms the third: the realization branch – where we realize our system. The diagram below illustrates the branches of 2TUP.

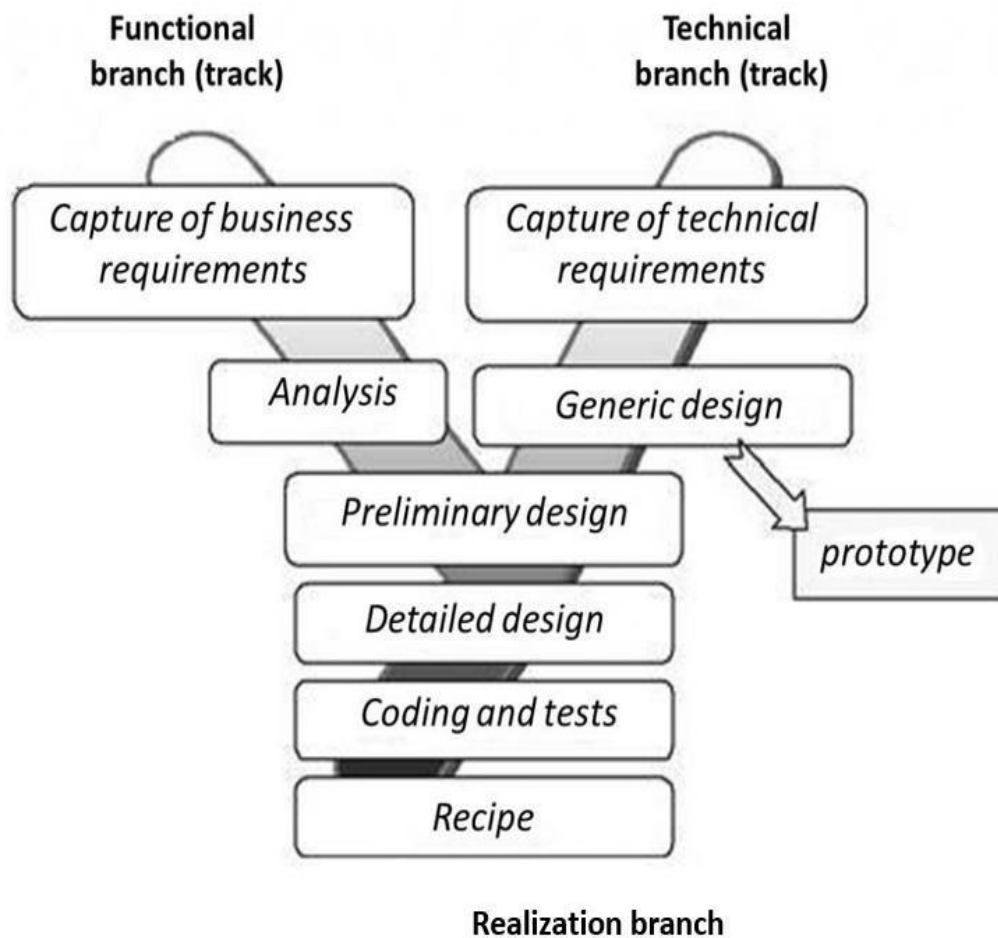


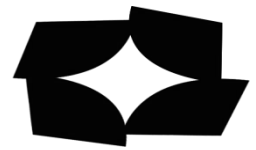
Figure 3: Two Track Unified Process (source: <https://www.mysciencework.com/omniscience/pervasive-mobile-healthcare-system-basedon-cloud-computing>)

a) The left branch (functional branch)

It captures the functional needs of a system. This ensures the production of software that meets the needs/requirements of the user. The analysis here consists of studying precisely the functional specification in order to obtain an idea of what the system is going to realize, and its result does not depend on any technology.

b) The right branch (Technical branch)

The technical branch enumerates the technical needs and proposes a generic design validated by a prototype. The technical needs include constraints and choices related to the conception of



the system, the tools and equipment as well as the integration constraint with the existing system condition.

c) The middle branch (Realization or Implementation branch)

In this branch, we study the preliminary conception, detailed conception, and documentation of the system. The realization branch supports the following:

Preliminary conception: This is the most sensitive step of 2TUP as it is the confluence of the functional and technical branch. It is completed when the deployment model, the operating model, the logical model, interphases and the software configuration model are defined. We have the following diagrams:

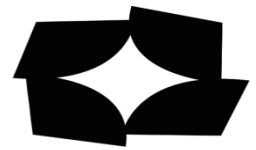
- Component Diagram
- Deployment Diagram
- Package Diagram
- Composite Structure Diagram

Detailed conception: This is the detailed design of each feature of the system. We have the following diagrams:

- Class diagram
- Object diagram
- Sequence diagram
- Timing Diagram

Coding and testing: This is the phase where we program the designed features and test the coded features.

The recipe: Also known as the deliverables is the validation phase of the functions of the developed system.



III. JUSTIFICATION OF THE ANALYSIS METHOD

The reason why we chose UML modelling language and the software development process 2TUP instead of many others that exist, include:

- UML is the current standard for programming in an object-oriented language. For this reason, it is widely understood and well known making it easy for a new programmer to join the project and be productive from the very first day.
- UML diagrams allow teams to virtualize how a project is or will be working, and they can be used in any field, not just software engineering. The diagrams will allow teams to virtualize together how a system, or a process will work or did work. It can provide new ideas for how teams have to collaborate to achieve the goal of the workflow process.
- 2TUP is centered around the creation and maintenance of a model, rather than the production of mountain documents.
- 2TUP is user oriented as it permits the development of software that responds to the needs of the users through the study of the user needs.
- 2TUP is iterative and incremental, hence it enables the project team to produce refined amelioration if necessary and easily integrate it in the already existing system.
- 2TUP by permitting the project team identify and test the key functionalities of the system limits the risk related to building the system.

IV. MODELLING OF THE PROPOSED SOLUTION

A. Capture of Functional Needs

The first step of the left (functional) branch of Two Track Unified Process (2TUP) is the capture of the functional needs. At this step, we capture the intended behavior of the system that maybe express as services, tasks or functions the system is required to perform.

1. USE CASE DIAGRAM

a. Definition

Use case diagram shows the functionalities of a system, their interdependencies and how they relate with actors of the system. A use case is a specification of behavior. The main objectives of the use case diagram are:

- Provide a high-level view of the system.
- Identify the functions of the system.

Use case diagrams are completed with a textual description of each use case that is intended to define the use case in details.

b. Formalism

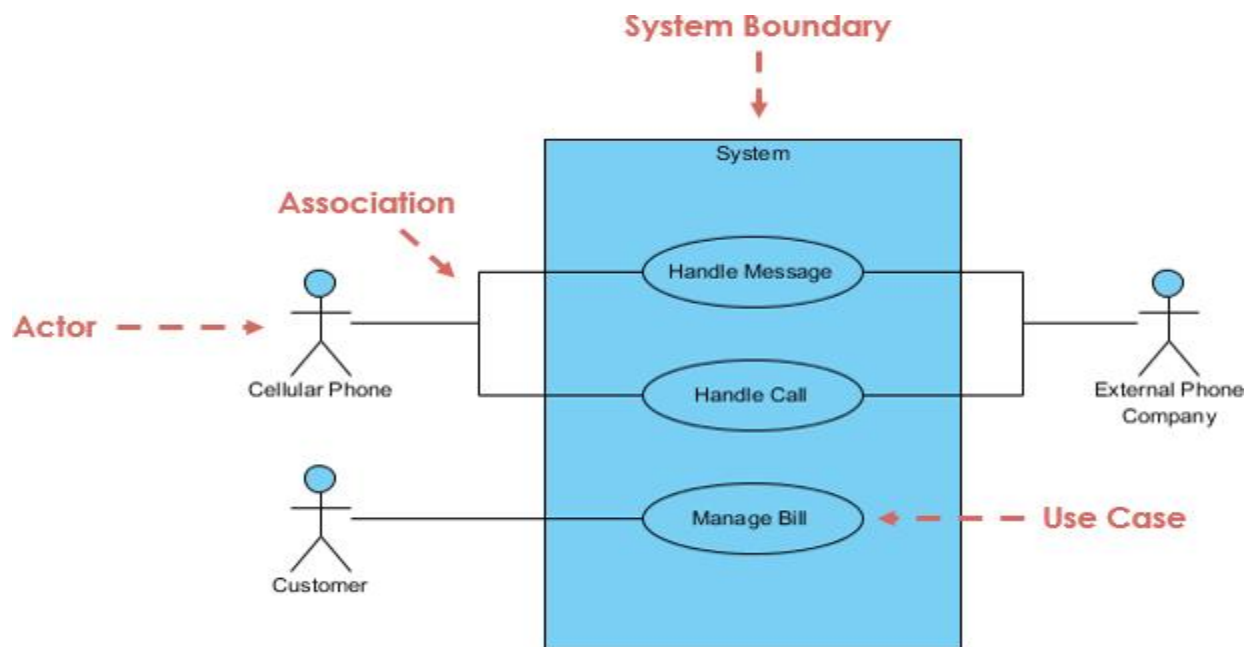
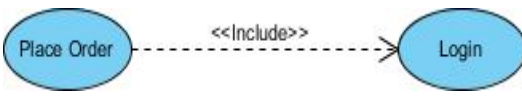
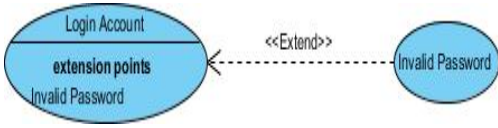
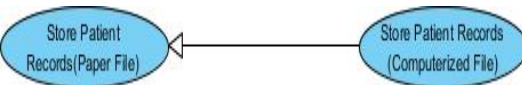


Figure 4: Use case diagram formalism

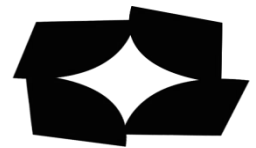
c. Components of use case diagram

Table 11: Component of use case diagram

Element	Notation	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor.
Association		it indicates that an actor takes part in a use Case
Include		An inclusion denotes that an included action must be performed before the including action can be performed.
Extend		An extension denotes that an extending action may be performed while an extended action is being performed.
Generalization		This shows that an actor or a use case is a kind of another abstract or concrete actors which can be defined and later specialized using generalization relationship.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON

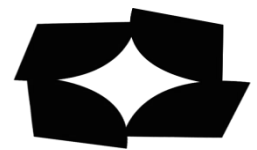


System		It is a container of use cases which interact with external actors
--------	---	--

d. Actors of our system

Table 12: Actors of our system

Actor	Role
Visitor	• Create an account • Browse course list
Student	Register for a course, View course, View lecture, View course progress, Take exercise, View exercise solution, Make payment, Make payment with OM (Orange Money) , Make payment with MOMO (Mobile Money) , Authenticate (for secure access), Access premium functionality, View past questions, View past questions & answers, View live practical
Teacher	Manage course material, Add course material, Update course material, Delete course material, Manage exercises, Add exercise, Update exercise, Delete course exercise, Make suggestion
Admin	Manage course, Add course, Update course, Manage user account, Add user account, Delete user account



e. General use case diagram

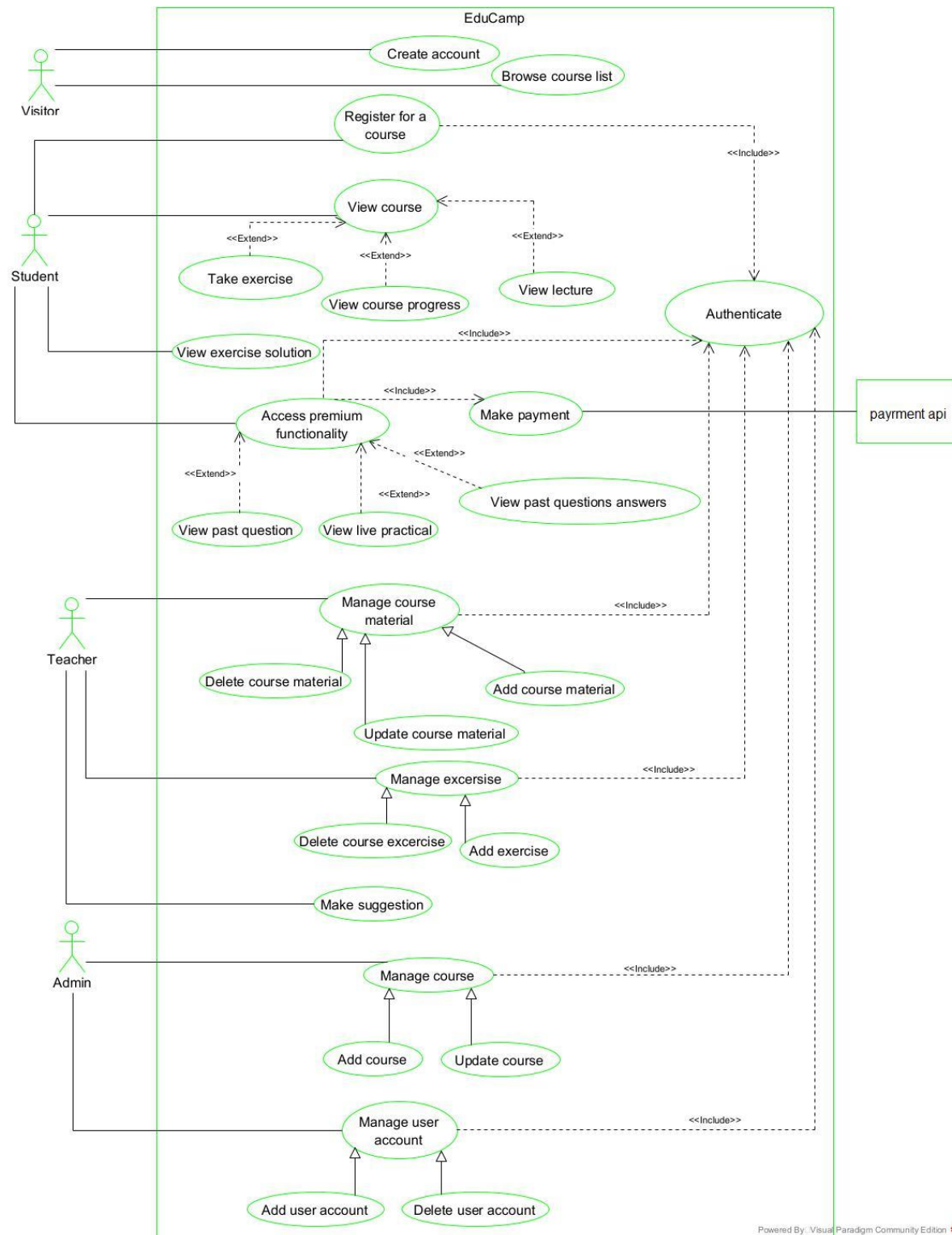


Figure 5: General use case diagram



F. Make Payment specific use case diagram

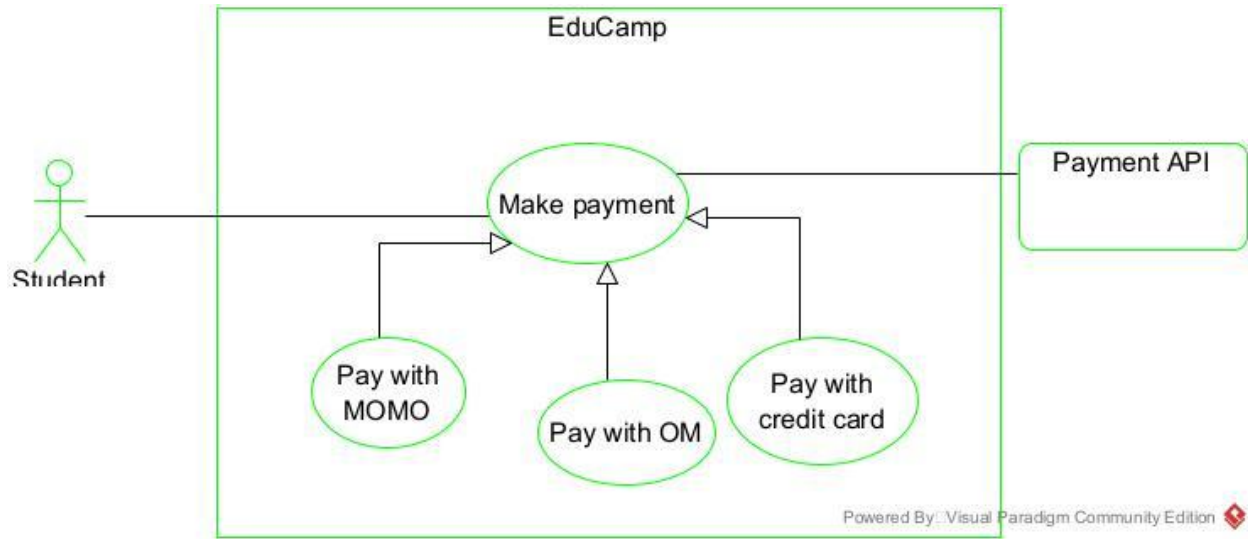
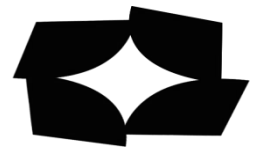


Figure 6: Make Payment use case diagram



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



Textual Description

UML allows the execution of a use case to be described in a textual way, in a form called nominal scenario. A nominal scenario describes in more detail the execution of a use case by an actor until it is successfully completed.

Formalism

A textual description of a use case is represented in the following form

Table 13: Textual description formalism

NAME OF THE USE CASE	
Summary	Aim of the use case
Actors	Users
Pre-condition	Set of actions that must be completed before the launching of the use case
Trigger	The event that causes the use case to be initiated
Nominal scenario	It corresponds to a normal development of a use case
Alternative scenario	These are variants of the nominal scenario.
Post condition	Set of mechanism that can lead to the end of the use case

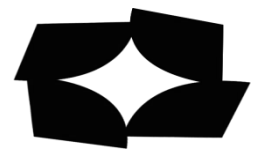


Table 14: Textual description of Authenticate

USE CASE AUTHENTICATE	
Summary	Permit client to access the system
Actors	Student, teacher, Admin
Pre-condition(s)	- The actor already has an account; - The app is launched - The account must be active
Trigger	The user clicks on login button
Nominal scenario	- The user clicks on login button - The system displays the login page - The user fills form and submit - The system checks conformity - The system sends validation query - The system treats the result returned by the DBMS - The user is directed to his page
Alternative scenario	- If step 4 of the nominal scenario the format is not correct, an error message is returned to the user and he returns to step 3 of the nominal scenario. - If step 6 of the nominal scenario no user was found, the system returns an error message and returns to step 3 of the nominal scenario
Post-condition of success	The user logs in successfully
Post condition of failure	An error message is display and user does not have access his Space.

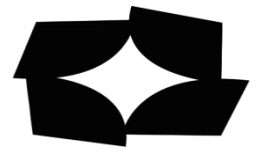


Table 15: Textual description of Create account

USE CASE CREATE ACCOUNT	
Summary	Grant the users access to the application
Actors	Visitor
Pre-condition(s)	The app is launched
Trigger	The user clicks on create button
Nominal scenario	- Click on create account - The system displays create account form - The user fills form and submit - The system checks for conformity of the form - The system sends a create account query - The system treats the result returned by the database - The system displays successful message
Alternative scenario	- If step 4 of the nominal scenario the format is not correct, an error message is returned to the user and he returns to step 2 of the nominal scenario. - If step 6 of the nominal scenario an error is found, an error message is returned to the user and he returns to step 2 of the nominal scenario
Post-condition of success	The user's account is created
Post condition of failure	An error message is displayed

2. ACTIVITY DIAGRAM

a. Definition

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process; showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended to model both the computational and organizational process. They flow can be sequential, branched or concurrent. Below is an activity diagram formalism.

b. Formalism

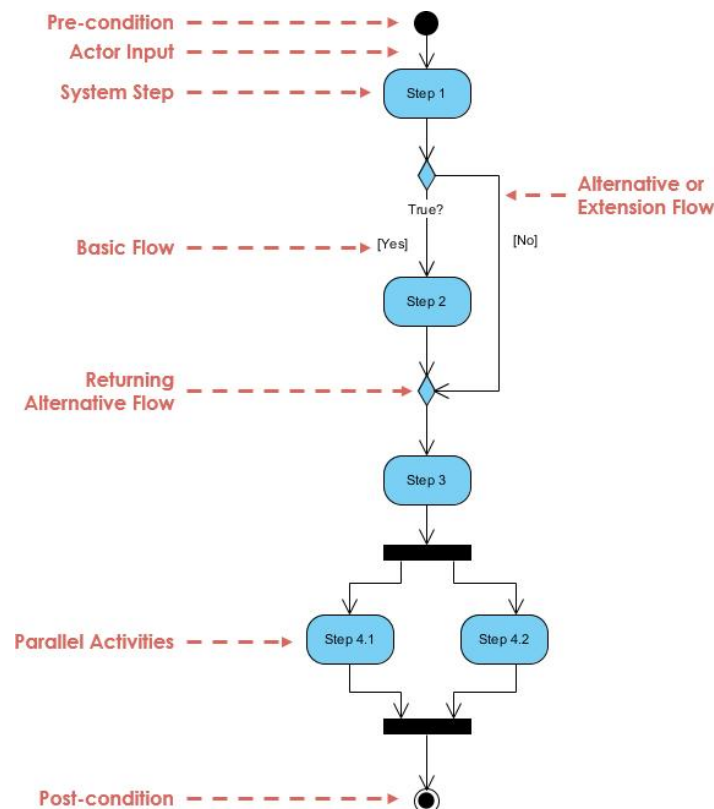
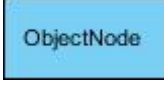
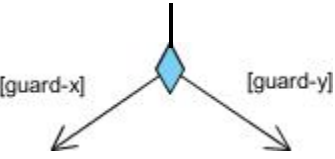
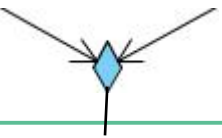


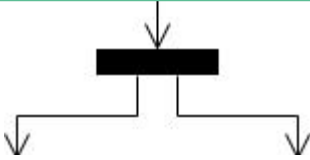
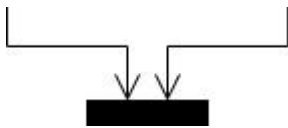
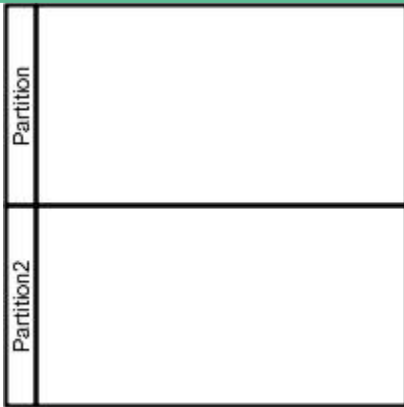
Figure 9: Activity diagram formalism

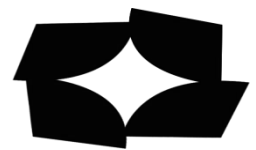
c. Component of an Activity diagram

Table 16: Component of an Activity diagram

Element	Notation	Description
Activity		Is used to represent a set of actions
Action		A task to be performed
Activity Edge		An activity edge in a UML activity diagram is a directed connection between two activity nodes. It represents the flow of control or data between these nodes.
Initial node		Portrays the beginning of a set of actions or activities
Activity Final Node		Stop all control flows and object flows in an activity (or action)
Object node		Represent an object that is connected to a set of Object Flows
Decision node		Represent a test condition to ensure that the control flow or object flow only goes down one path
Merge node		Bring back together different decision paths that were created using a decision-



		node.
Fork node		Split behavior into a set of parallel or concurrent flows of activities (or actions)
Join node		Bring back together a set of parallel or concurrent flows of activities (or actions).
Swimlane and Partition		A way to group activities performed by the same actor on an activity diagram or to group activities in a single thread



d. Activity diagram of Authenticate

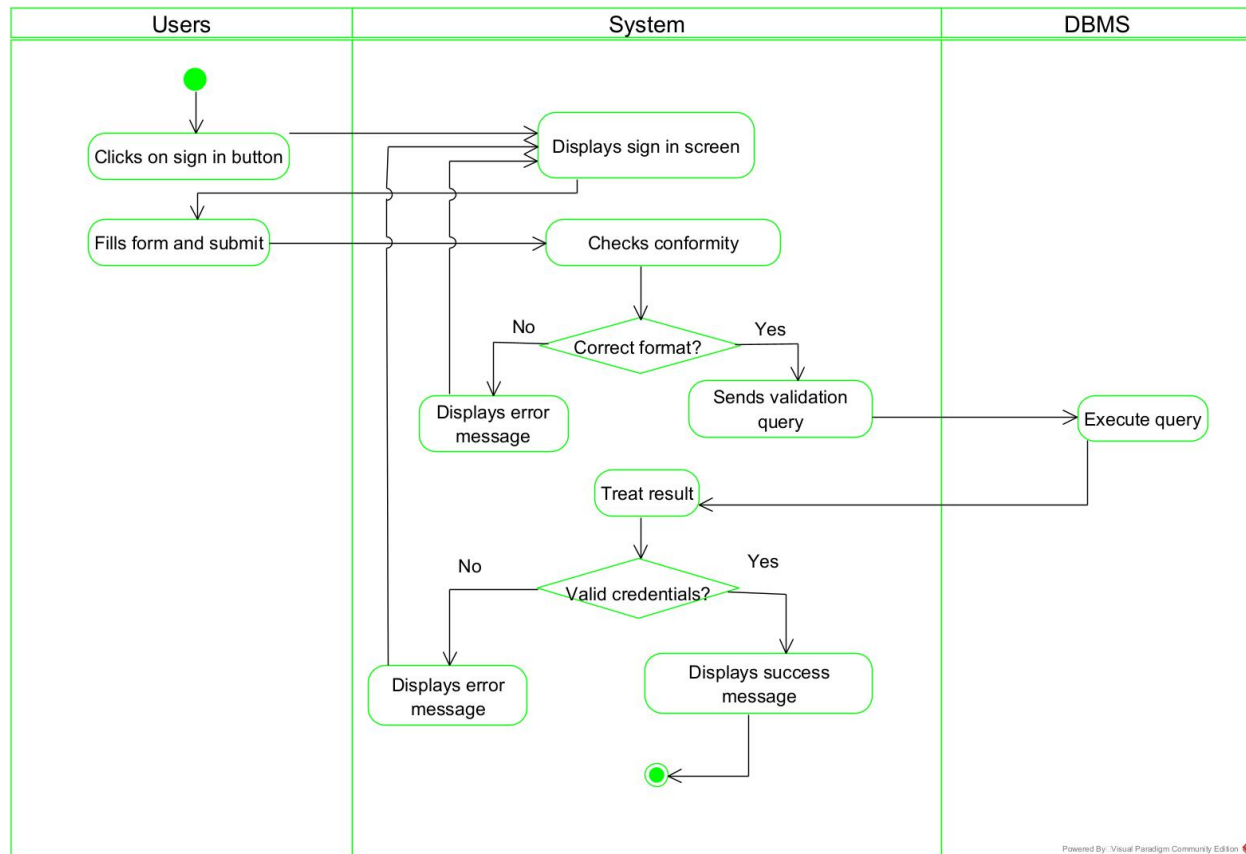
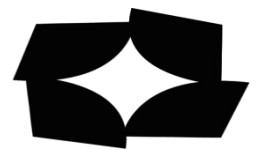


Figure 7: Authenticate activity diagram



e. Activity diagram of Create account

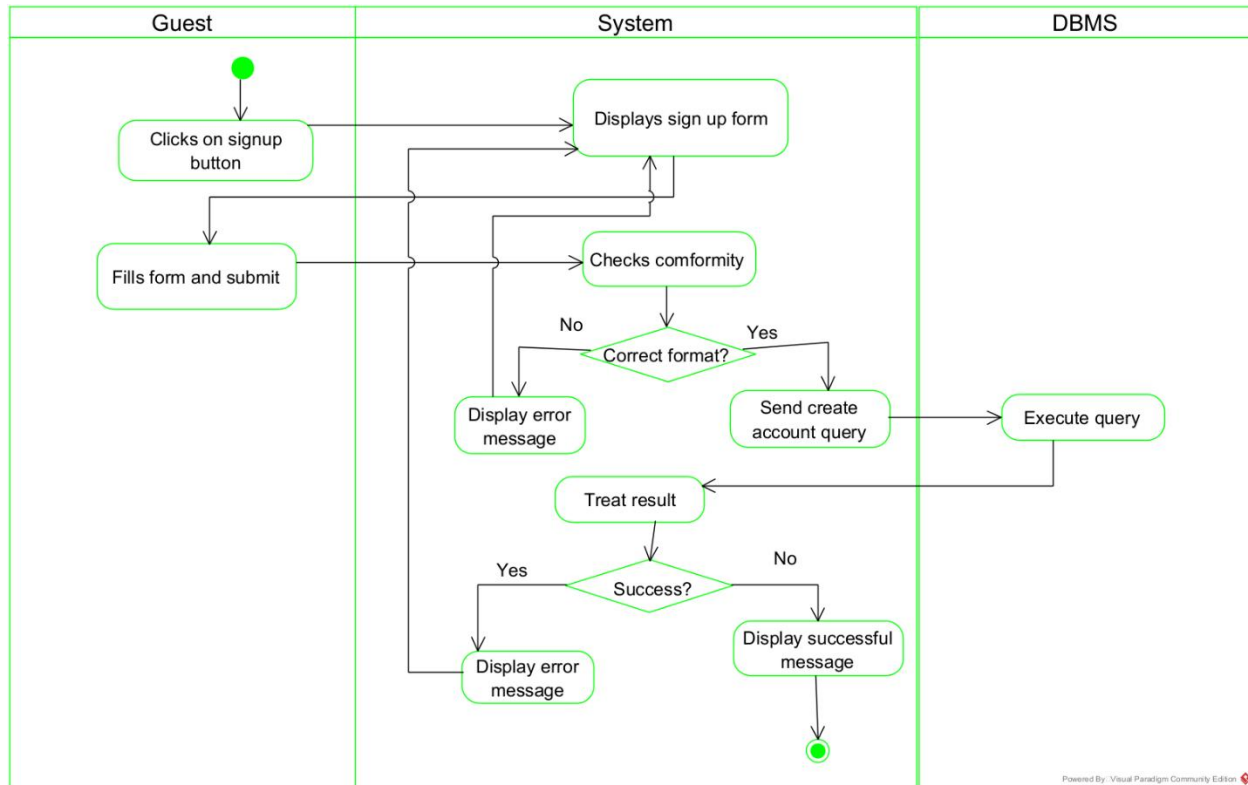


Figure 8: Create account activity diagram



f. Activity diagram of Make Payment

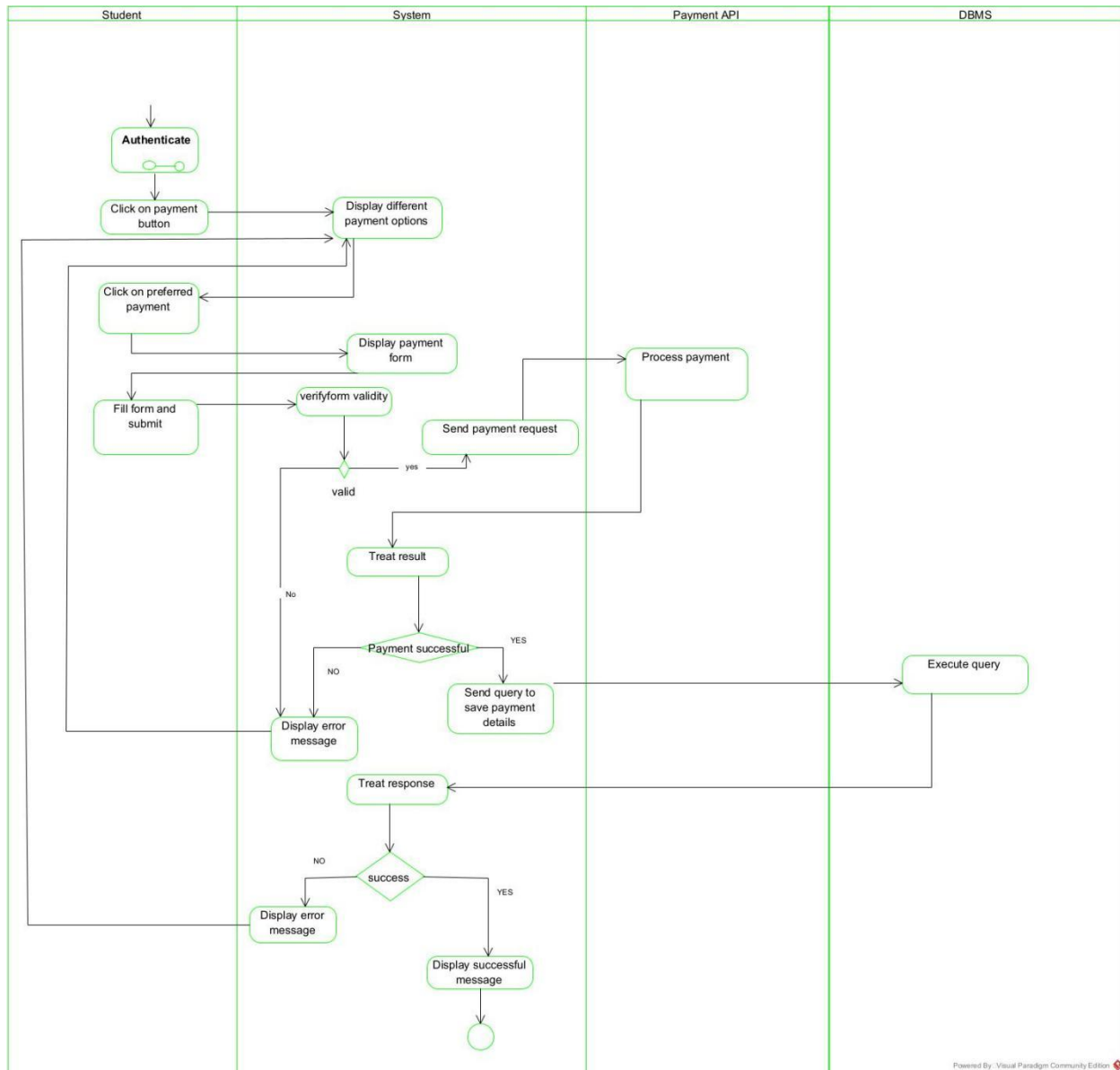


Figure 9: Make Payment activity diagram

3. COMMUNICATION DIAGRAM

a. Definition

Communication Diagrams model the interactions between objects in a sequence. They describe both the static structure and the dynamic behavior of a system. It is a simplified version of a Collaboration Diagram introduced in UML 2.0. A communication diagram is more focused on showing the collaboration of objects rather than the time sequence.

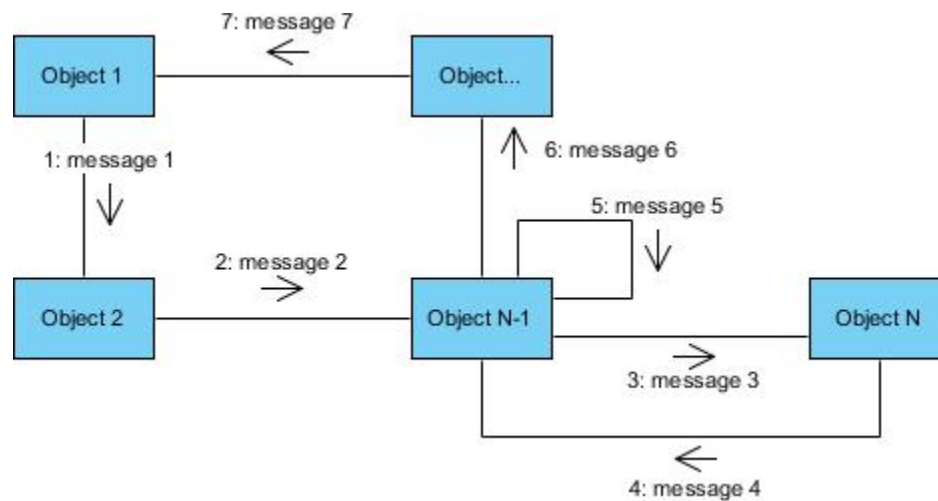
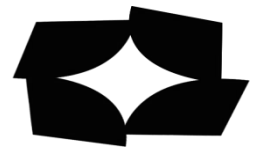


Figure 14: Communication diagram formalism

b. Formalism



c. Components of a communication diagram

Table 17: Component of a communication diagram

Element	Notation	Description
Message	2: message 2 →	Designs a particular communication between lifelines.
Actors		A role play by an entity that interacts with the subjects
Link		It initiates an association it connects two objects together for them to communicate.
Object		An actor represents an individual participant in the interaction conversation

d. Communication diagram of Authenticate

2.1: Treat result

2.2: Send authentication query

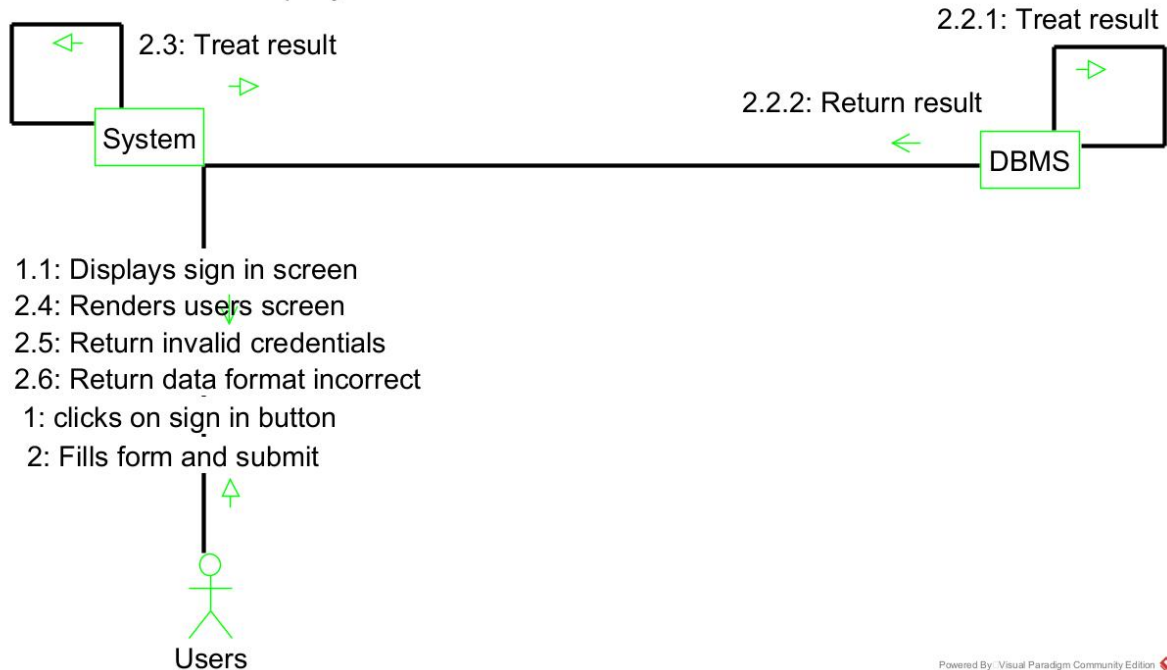
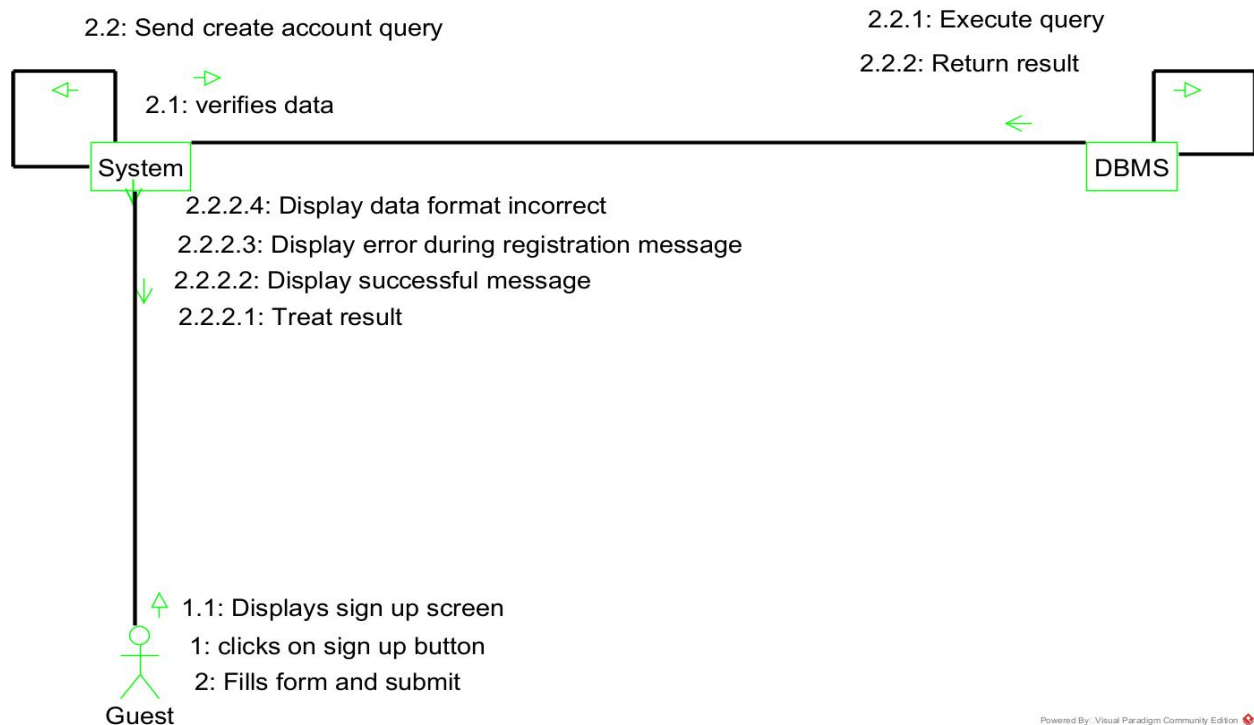
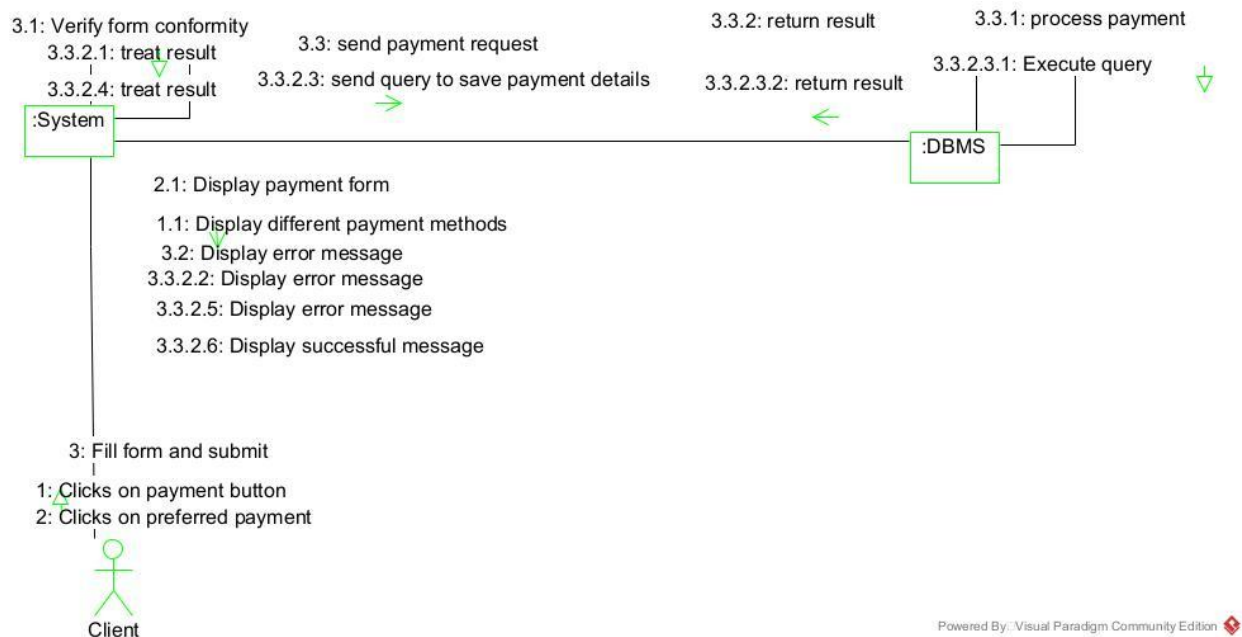


Figure 10: Authenticate Communication diagram

e. Communication diagram of Create account



f. Communication diagram of Make Payment



4. SEQUENCE DIAGRAM

a. Definition

A Sequence diagram describes interactions among classes in terms of an exchange of messages over time. They are also called event diagrams. A Sequence diagram is a good way to visualize and validate various runtime scenarios. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modelling a new system.

b. Formalism

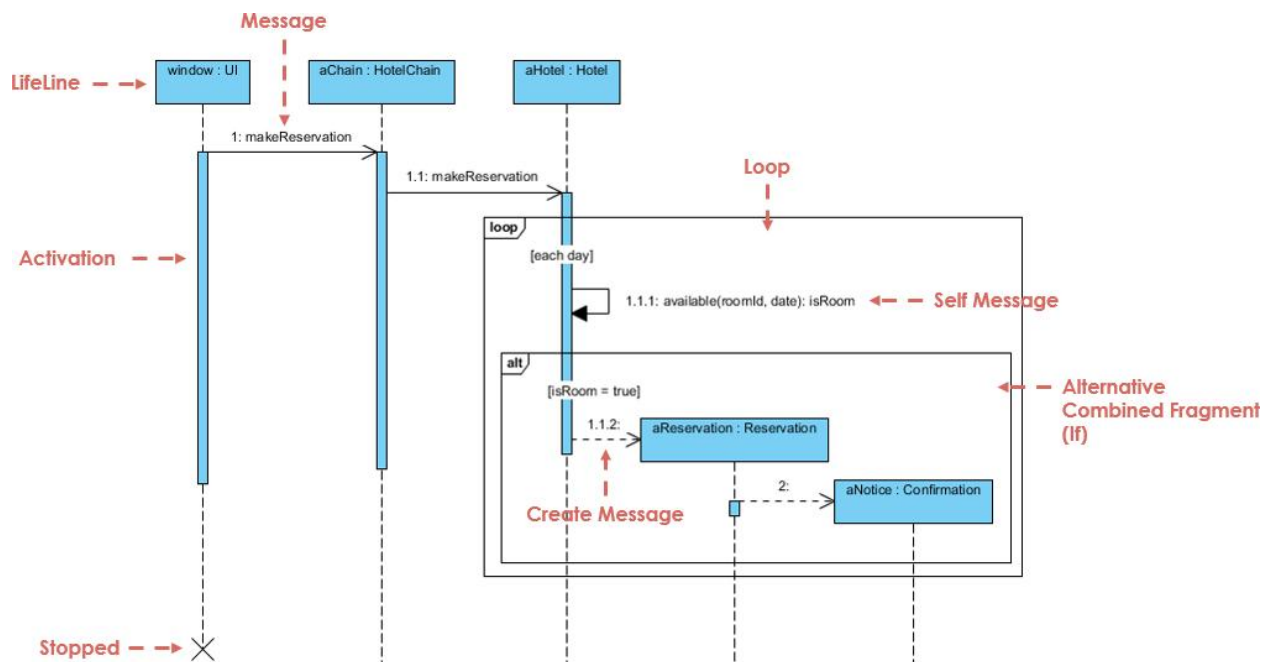
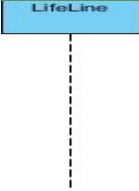
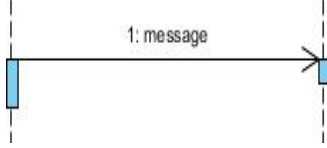
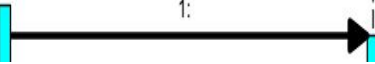
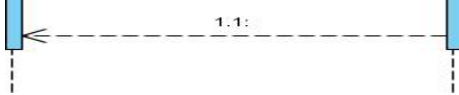
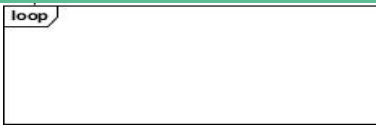


Figure 13: Sequence diagram formalism

c. Component of sequence diagram

Table 18: Component of sequence diagram

Element	Notation	Description
Lifelines		They represent rows or objects instances that participate in the sequence being modelled.
Asynchronous Message		It is a message that receives an indirect response.
Synchronous Message		It is a message that sends and want response before it continues a process
Self-Message		Self message is a kind of message that represents the invocation of message of the same lifeline.
Return Message		It represents the response of a message.
Actor		They send and receive message.
Combined Interaction Fragment		An articulation of interaction diagram, defined by an operator and operands.
Object		They send and receive messages

d. Sequence diagram of Authenticate

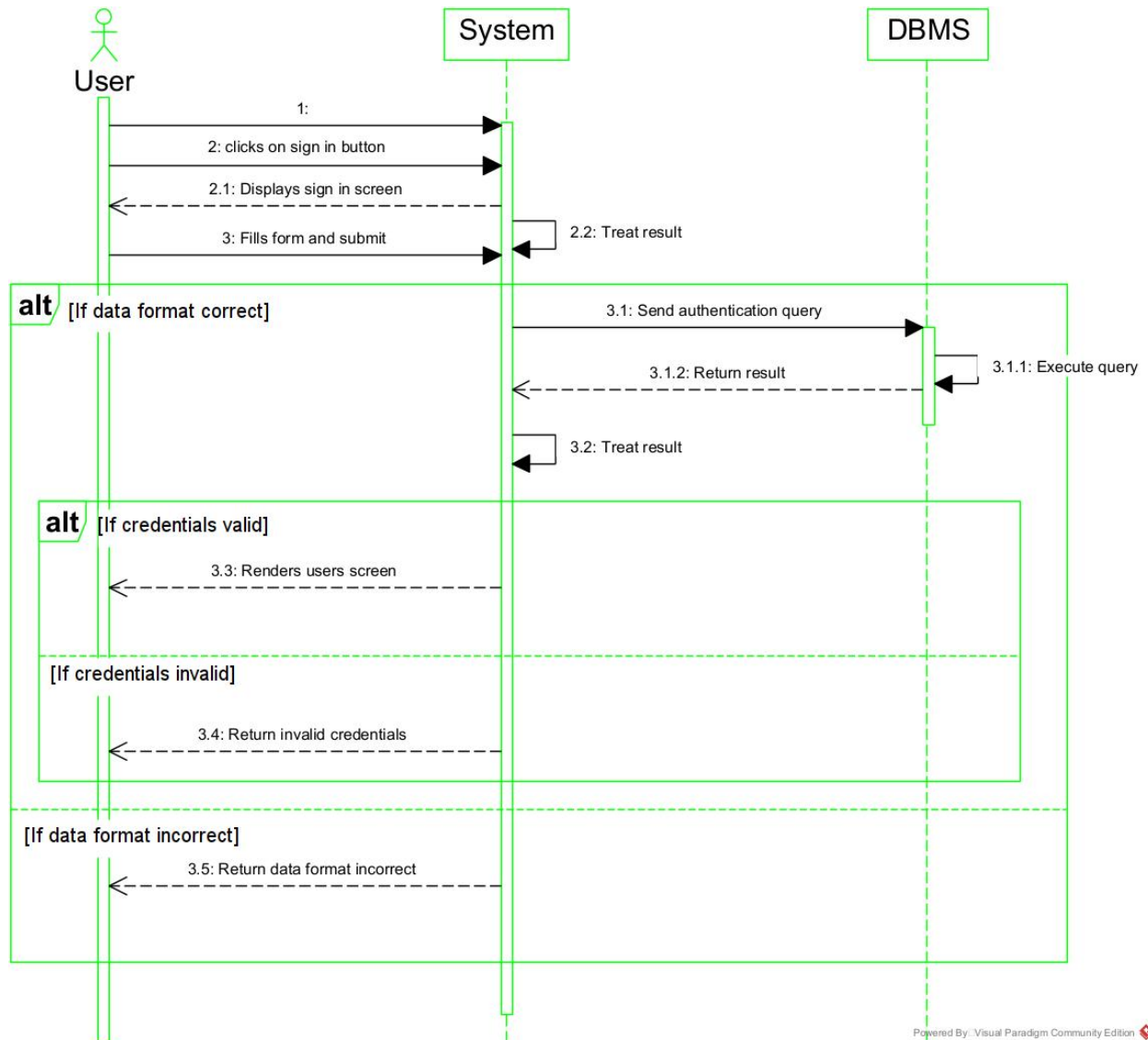


Figure 14: Authenticate Sequence diagram



e. Sequence diagram of Create account

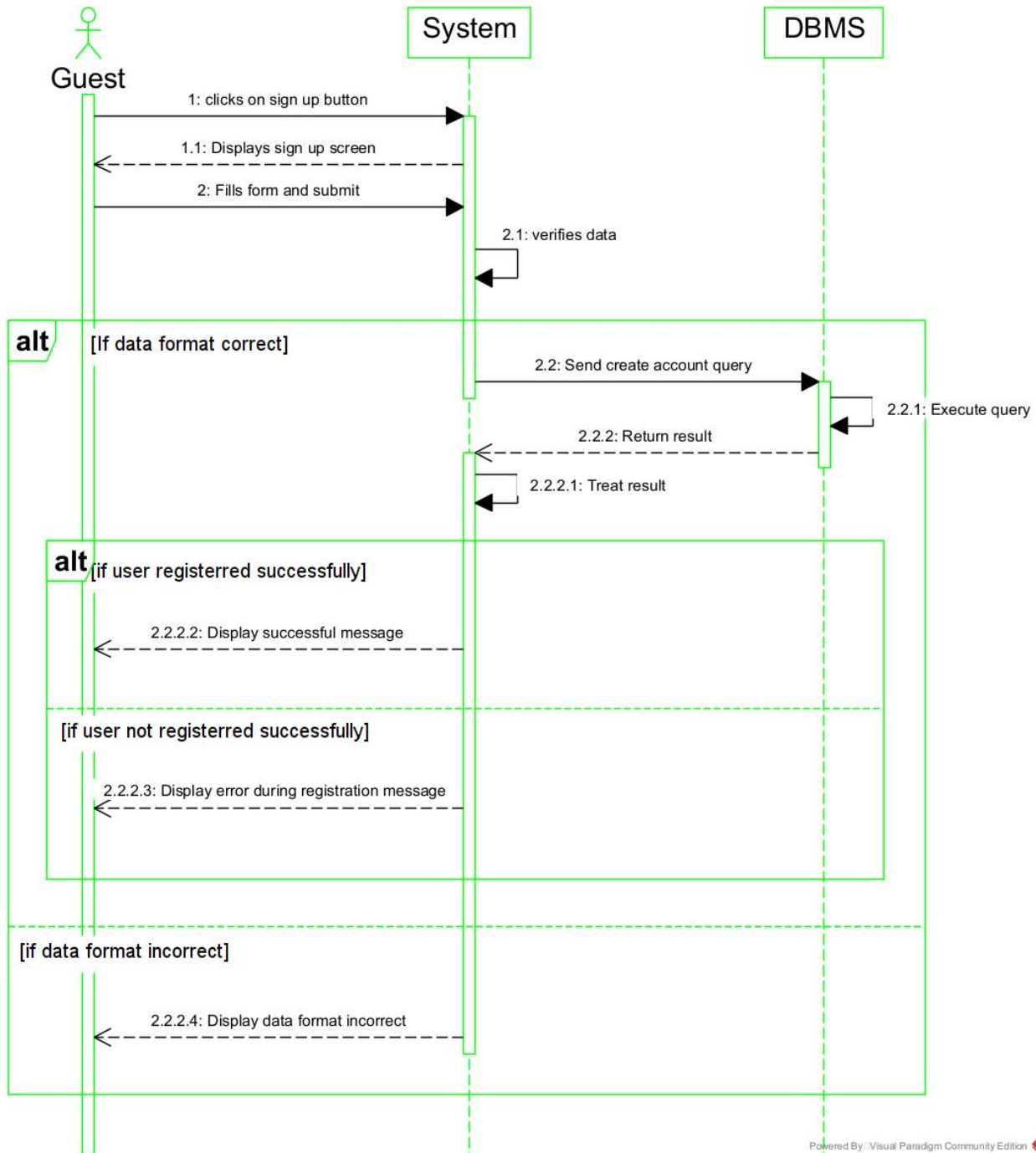


Figure 15: Create account sequence diagram

f. Sequence Diagram of Make Payment

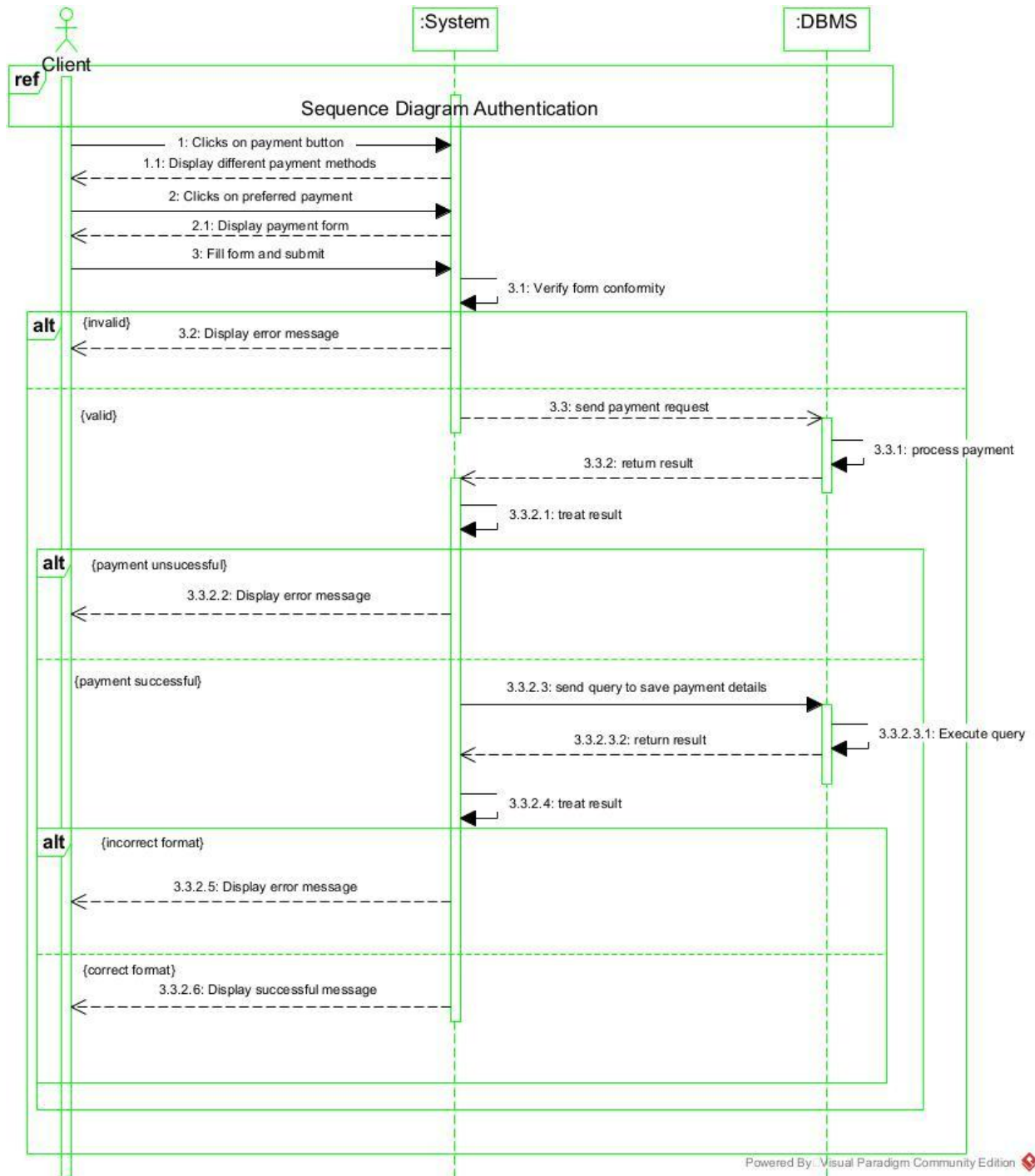
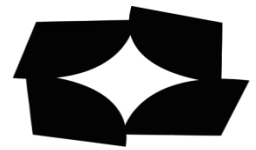


Figure 16: Create certificate sequence diagram

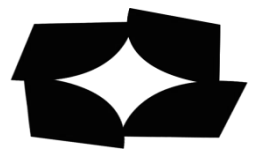


CONCLUSION

During the analysis phase, we selected a software development process and modelling language. Next, we elaborated on the functional requirements of our system. We reviewed the use case diagram, which outlines the relationship between actors and the actions they can perform in the system, as well as the communication diagram that illustrates the architecture of the system based on object-oriented programming. We also analysed the sequence diagram, which delineates the flow of messages between elements in the system. Lastly, we reviewed the activity diagram, which displays the workflow of our system. We will proceed to the conception phase, during which we shall showcase the Technical branch of our system, accompanied by relevant diagrams.



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



DOCUMENT IV: CONCEPTION PHASE



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



PREAMBLE

The conception phase will permit us to present in an orderly manner the components necessary for the good functioning of our software and also the architecture used for the proposed solution. It bridges the gap between the analysis phase and the realization phase.

CONTENT

INTRODUCTION

I TECHNICAL BRANCH

A. CAPTURE OF TECHNICAL NEEDS

B. RELATED UML DIAGRAMS

i. CLASS DIAGRAM

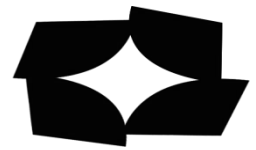
ii. STATE MACHINE DIAGRAM

iii. PACKAGE DIAGRAM

CONCLUSION



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



INTRODUCTION

The concept stage will outline the precise requirements, characteristics and activities needed to fulfil the operational needs of the suggested system, as defined during the analysis stage. The objective during this phase is to recognise and examine key components (either hardware or software), configuration (network capacity), processes and procedures which are indispensable for the system to achieve its desired outcomes. We will scrutinise several diagrams such as the class diagram, state machine diagram and package diagram.

I. TECHNICAL BRANCH

B. CAPTURE OF TECHNICAL NEEDS

a. Physical Architecture

The design of the DBMS depends on its architecture. An n-tier architecture partitions on the whole system into related but separated n modules, which can be independently modified, altered, changed or replace. A large amount of data on web servers, personal computers (pc) and others are link with networks with the help of basic client or server architecture.

Within the scope of our project, we made use of the 3-tier architecture This architecture separate it tiers from each other based upon the user and the manipulated data in the database. Each layer has a well-defined communication interface, and the evolution of the layer is independent of the other. The 3-tier of our system is made up of:

- ✚ The client tier, which is also known as our presentation interphase.
- ✚ Application Tier, which represents our webserver.
- ✚ The data tier, which represents our DBMS server

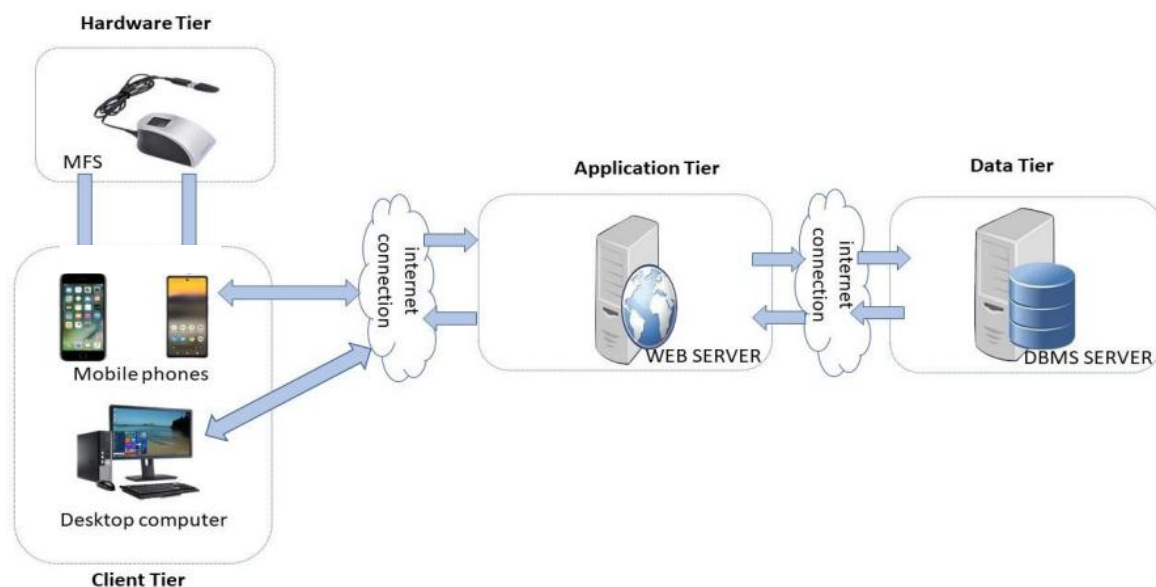


Figure 17: : n-tier Architecture (Source: <https://www.pinterest.es/pin/752241943987565648/>)



b. Logical Architecture

Model View controller or MVC as it is popularly called, is a software design pattern for developing application. A model view controller pattern is made up of the following three parts.

- ❖ **Model:** The lowest level of the pattern which is responsible for maintaining data.
- ❖ **View:** This is responsible for displaying all or a portion of data to the user.
- ❖ **Controller:** It handles software codes that controls the interactions between the model and the view.

MVC is popular as it isolates the application logic from the user interface and supports separation of concerns. Here the controller receives all requests for the application then works with the model to prepare data needed by the view. The view then uses the data prepared by the controller to produce a final response. The MVC can be represented as follows:

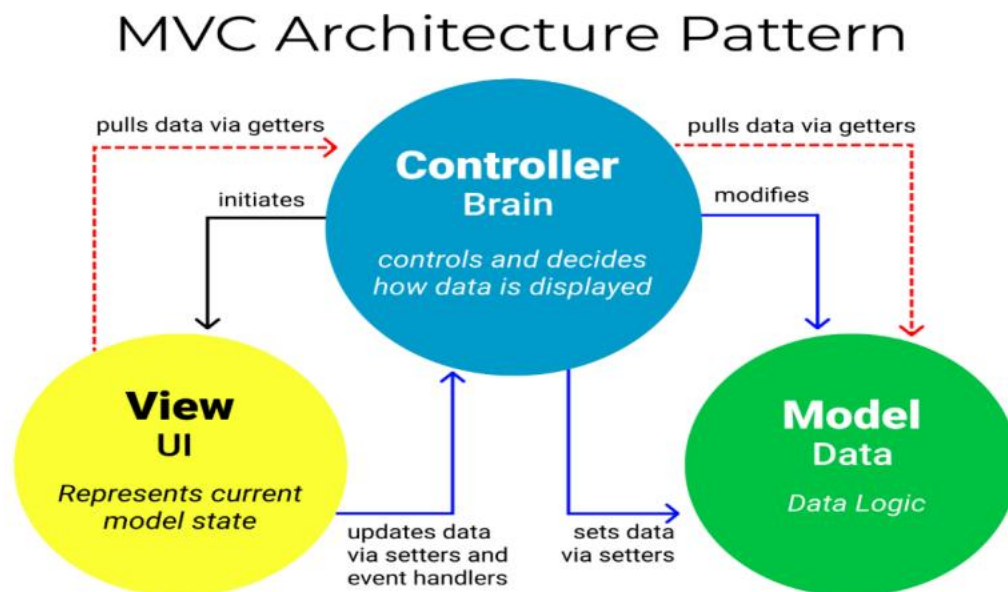


Figure 26: The MVC architecture (source: <https://www.freecodecamp.org/news/the-model-view-controller-pattern-mvc-architecture-and-frameworks-explained/>)



A. RELATED UML DIAGRAMS

1. CLASS DIAGRAM

a. Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

b. Formalism

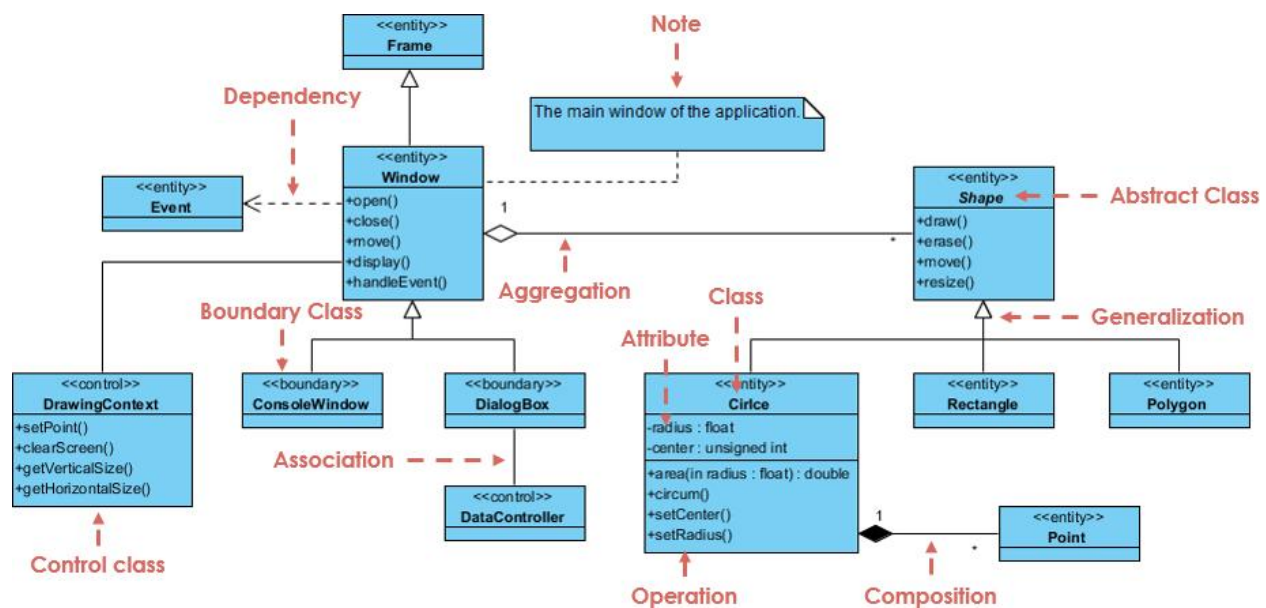
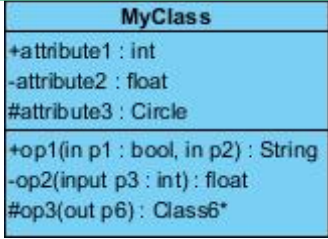
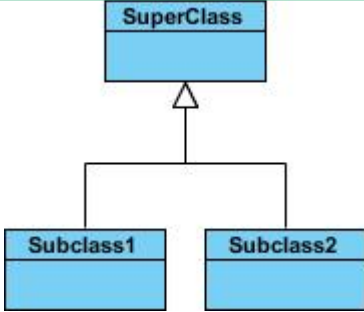
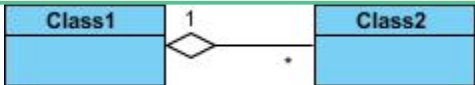
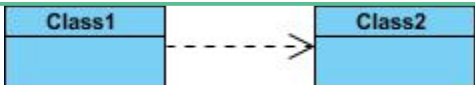


Figure 18: Class diagram formalism

c. Component of a Class diagram

Table 19: Components of class diagram

Element	Notation	Description
Class	 <pre> classDiagram class MyClass { +attribute1 : int -attribute2 : float #attribute3 : Circle +op1(in p1 : bool, in p2) : String -op2(input p3 : int) : float #op3(out p6) : Class6* } </pre>	A class is an element that defines the attributes and behaviours that an object can generate
Generalization	 <pre> classDiagram SuperClass < -- Subclass1 SuperClass < -- Subclass2 </pre>	it a relationship between a whole thing (called superclass) and a more specific thing (called subclass)
Association	 <pre> classDiagram Class1 -- Class2 </pre>	It is a general type of relationship between elements, it may include cardinality, roles etc.
Aggregation	 <pre> classDiagram Class1 o-- "1" Class2 </pre>	If the parent of the aggregate is deleted, the children are not deleted.
Composition	 <pre> classDiagram Class1 *-- "1" Class2 </pre>	If a parent of a composite is deleted all its parts are deleted with it.
Dependency	 <pre> classDiagram Class1 ..> Class2 </pre>	It existed between two classes, if one changes it may cause the change in the order, but the other way around

d. System Class diagram

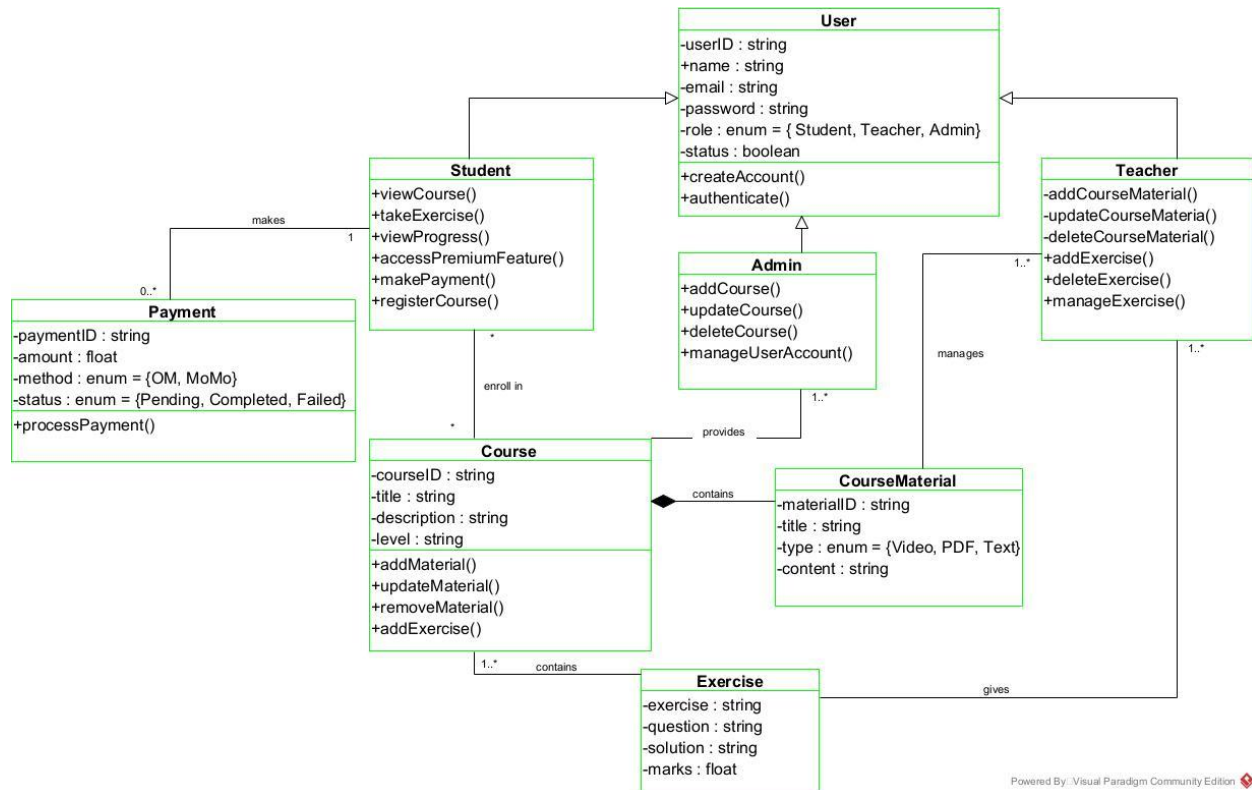


Figure 19: CamEdu Class diagram

2. STATE MACHINE DIAGRAM

a. Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

b. Formalism

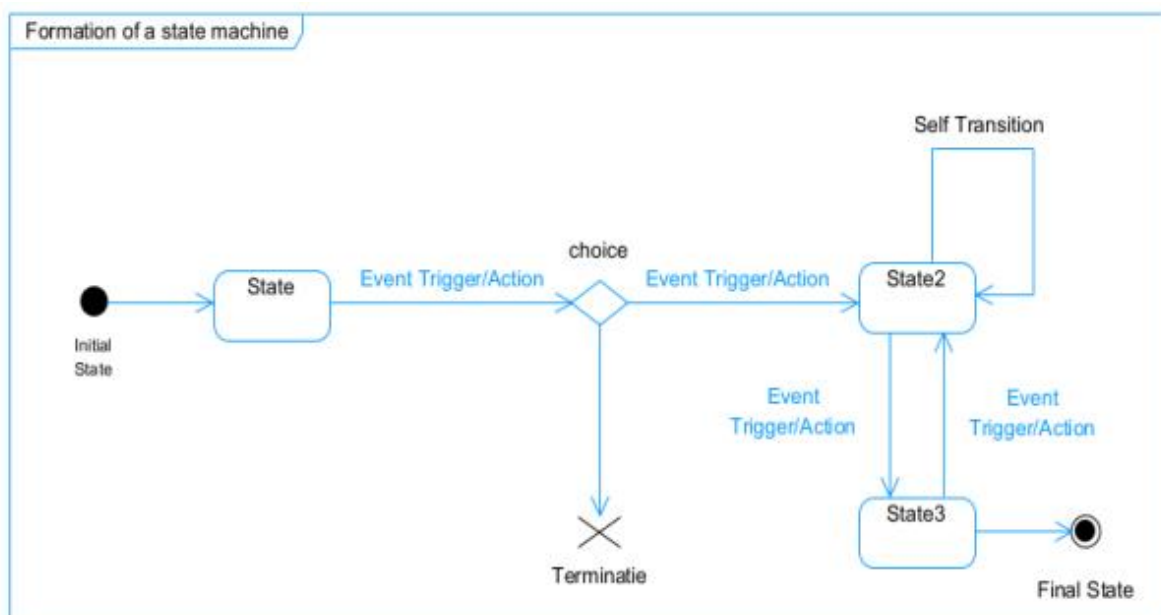
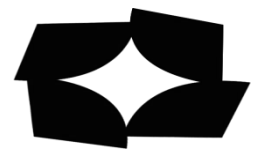
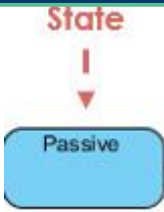
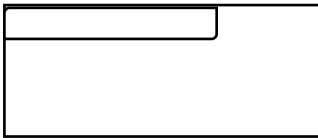


Figure 29: Formalism of a state machine diagram



c. Component of a state machine diagram

Table 20: Component of state machine diagram

Element	Notation	Description
State		Models a situation during which a certain invariant condition holds.
First (Initial State)		It represents a default vertex, that is, a source for a single transaction to the default or composite state.
Final state		A state specifying that the enclosing region is complete
Transition		A direction relation between a source and a target vertex.
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branched potential results
Terminate		Implies that the execution of a state by means of it context is terminated.
Diagram overview		A placeholder for the linked states in a state machine diagram.



d. State machine diagram of an account

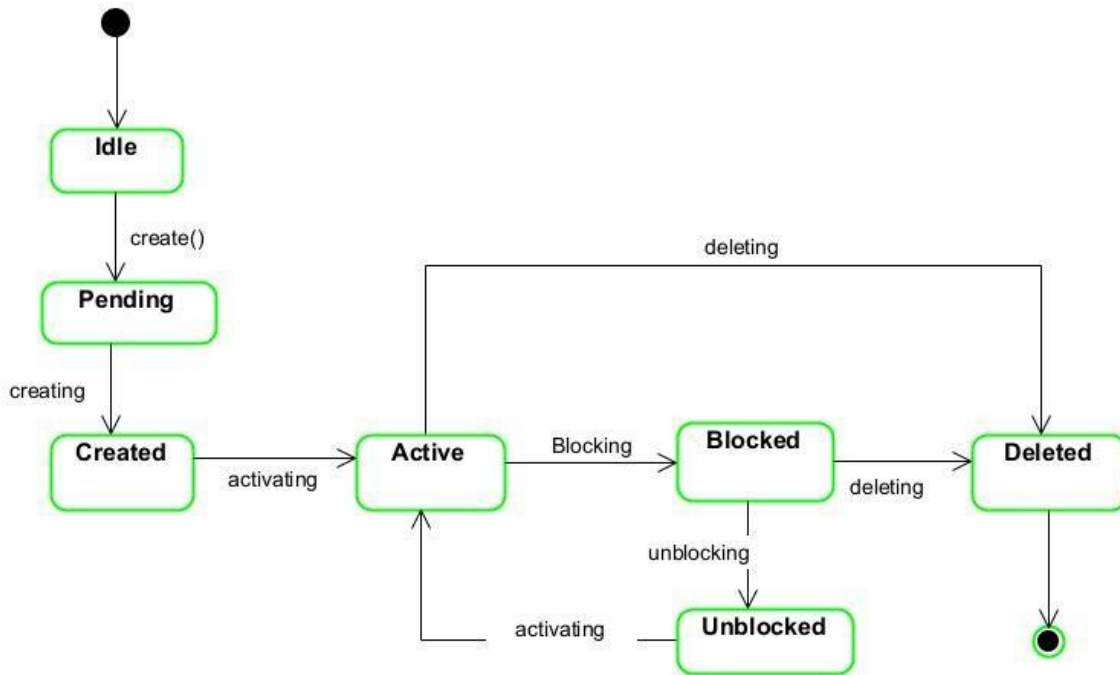


Figure 20: state machine diagram of an account



3. PACKAGE DIAGRAM

a. Definition

This is a structural diagram used to show the organization and arrangement of various model elements in the form of packages. A package diagram is the grouping of related uml elements such as classes, diagrams or eve other packages.

b. Formalism

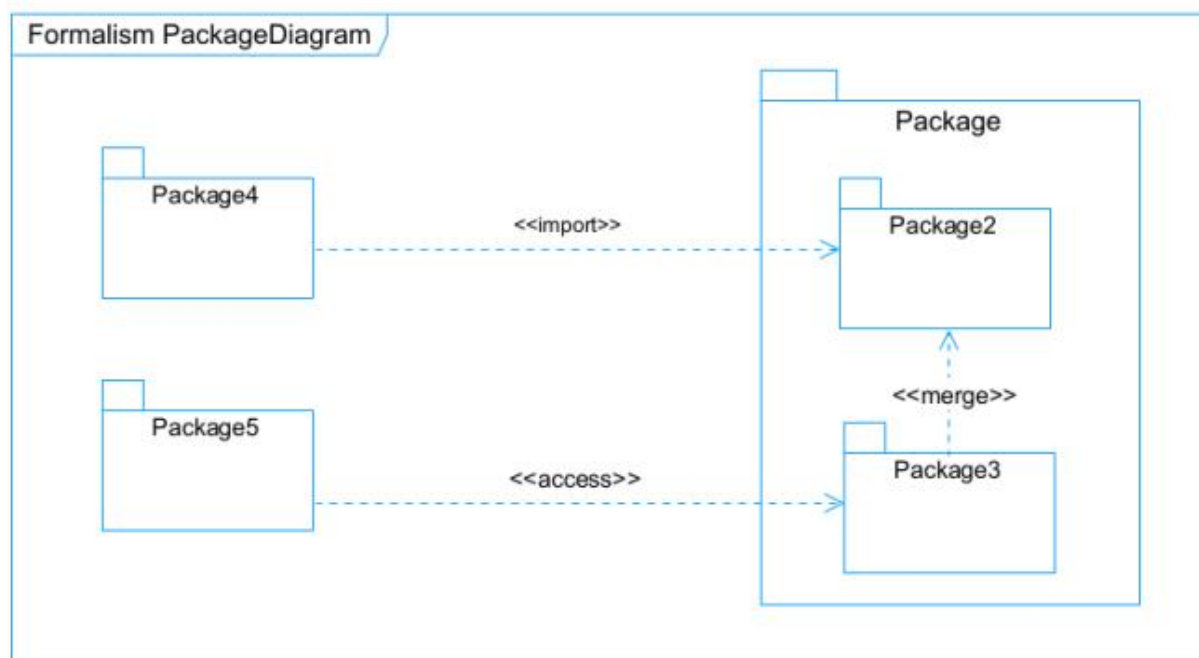
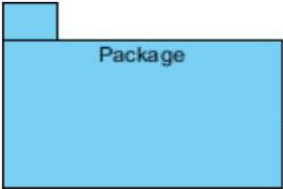
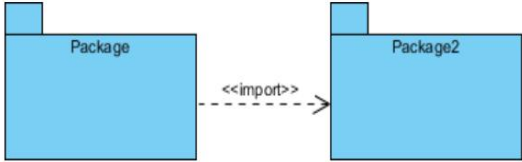
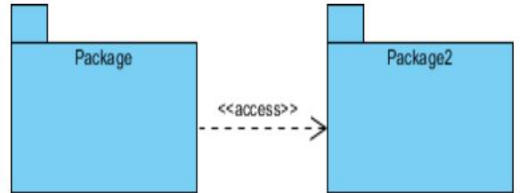


Figure 21: Package diagram formalism

c. Component of package diagram

Table 21: Component of package diagram

Element	Notation	Description
Package		A package is a namespace use to group related elements; it is a mechanism used to group elements into a better structure in a system.
Package import		A relationship Indicate that, functionality has been imported from one package to another.
Package access		A relationship Indicates that one package requires assistance from the function of another package.
Package merge		It is a relationship which shows that, the functionality of two packages are combines to a single function.



d. CamEdu Package diagram

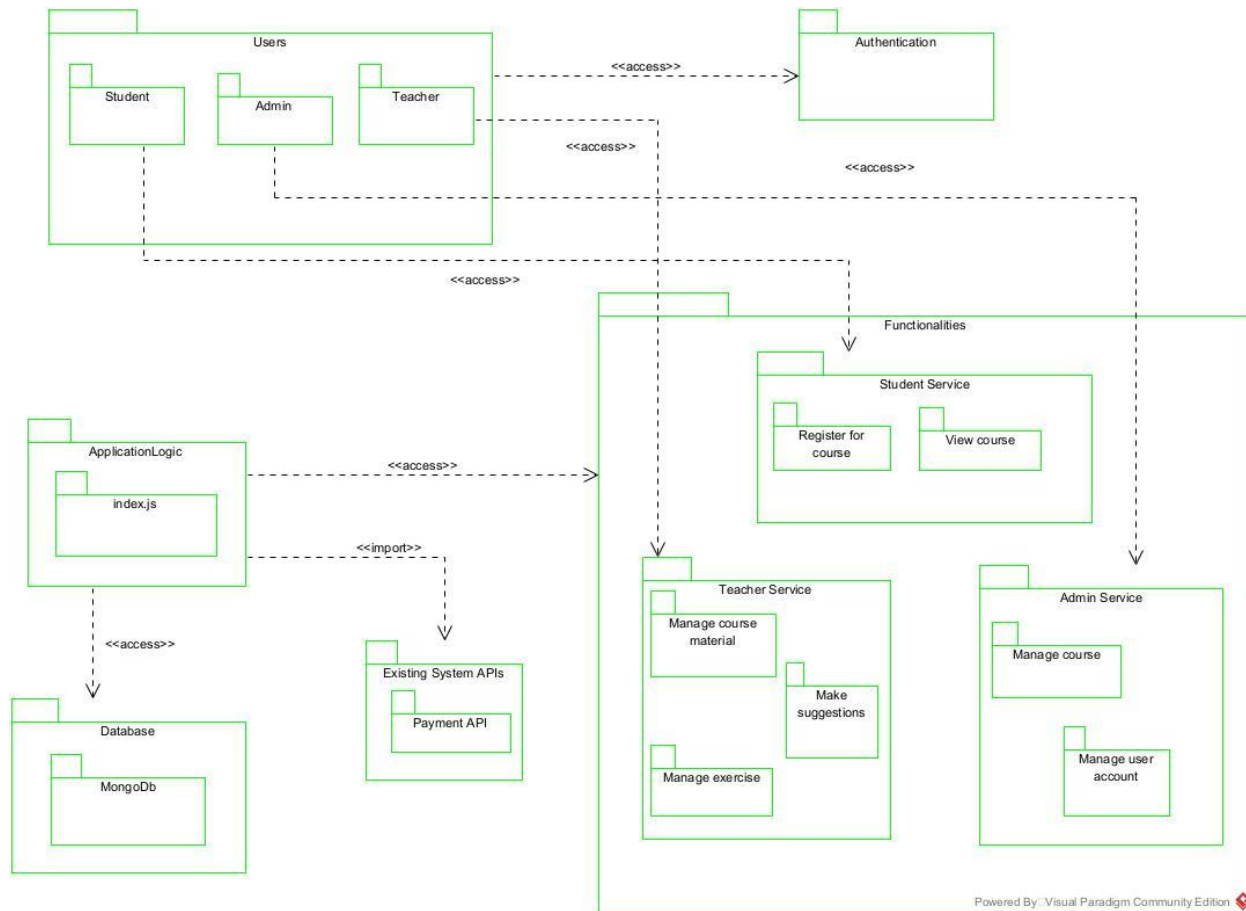
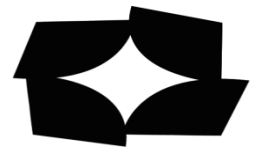


Figure 22: CamEdu package diagram



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON

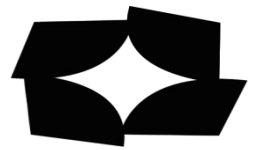


CONCLUSION

We presented the structure of our system and its integration into the existing technical architectures. In the initial stage, we aimed to investigate various aspects of our system. Primary actions included assessing the technical limitations, identifying the system's components with their classification and deployment onto relevant deployment targets. We then concluded this phase by evaluating the interactions between different components and participants of our system. Moving forward, our report will now progress to the implementation phase, which will entail concentrating on executing our system-related elements.



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DOCUMENT V: REALIZATION PHASE



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



PREAMBLE

In this phase, we aim to implement our solution straightforwardly by relying on the analysis and conception phases. Additionally, we will showcase the component and deployment diagrams.

CONTENT

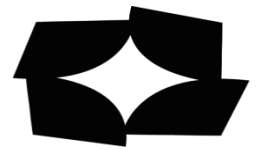
INTRODUCTION

- I. DEPLOYMENT DIAGRAM
- II. COMPONENT DIAGRAM

CONCLUSION



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



INTRODUCTION

Here in the realisation phase, we will review diagrams concerning the physical aspects of the system, such as libraries, documents and the physical topology of the system components when the software is deployed.



1. DEPLOYMENT DIAGRAM

a. Definition

Deployment diagram is a structural diagram used to visualize the topology of the physical components of a system, where the software is deployed. They consist of nodes and their relationship. It is related to the component diagram because the components are deployed using the deployment diagram. A deployment diagram consists of nodes. Nodes are nothing but physical hardware used to deploy the application.

b. Formalism

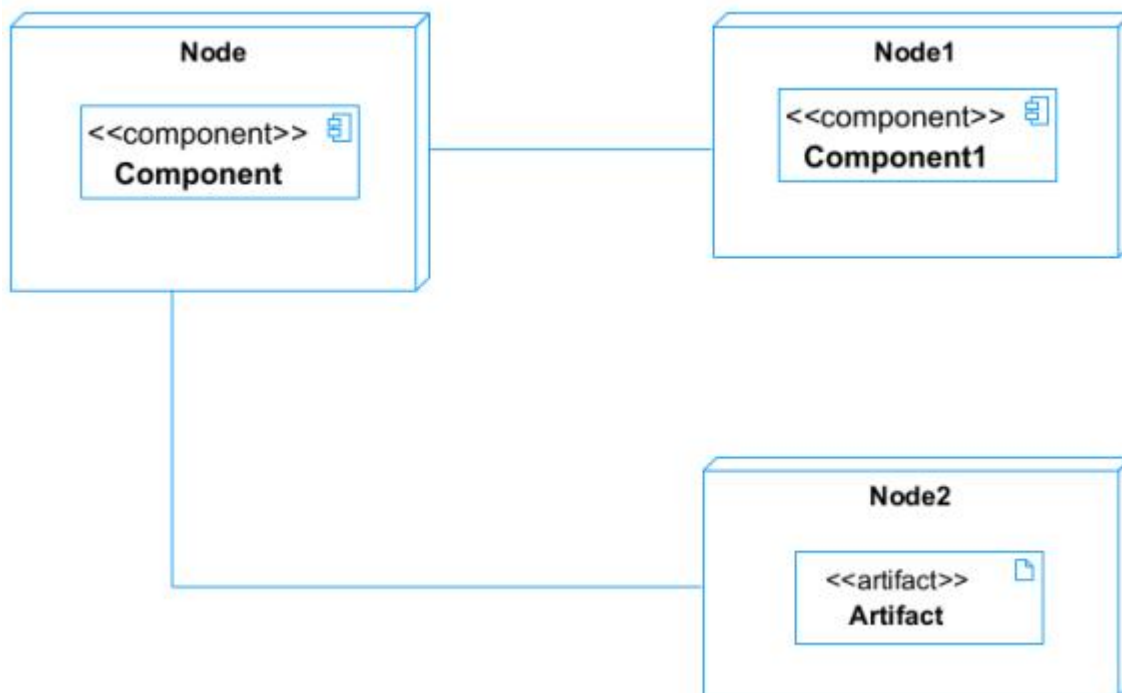
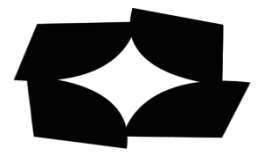
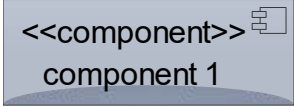


Figure 33: Formalism of De



c. Component of Deployment diagram

Table 22: Component of deployment diagram

Element	Notation	Description
Node		A hardware or software object, shown by a three-dimensional box
Artifact		An artifact is a major product, which is produced or used during the development of a software. E.g diagrams, data models, setup scripts
Component		It represents a modular part of a system that encapsulates its content and whose manifestation is replaceable within its environment.
Association		An association helps to connect two nodes together which permits them to communicate together



d. CamEdu Deployment diagram

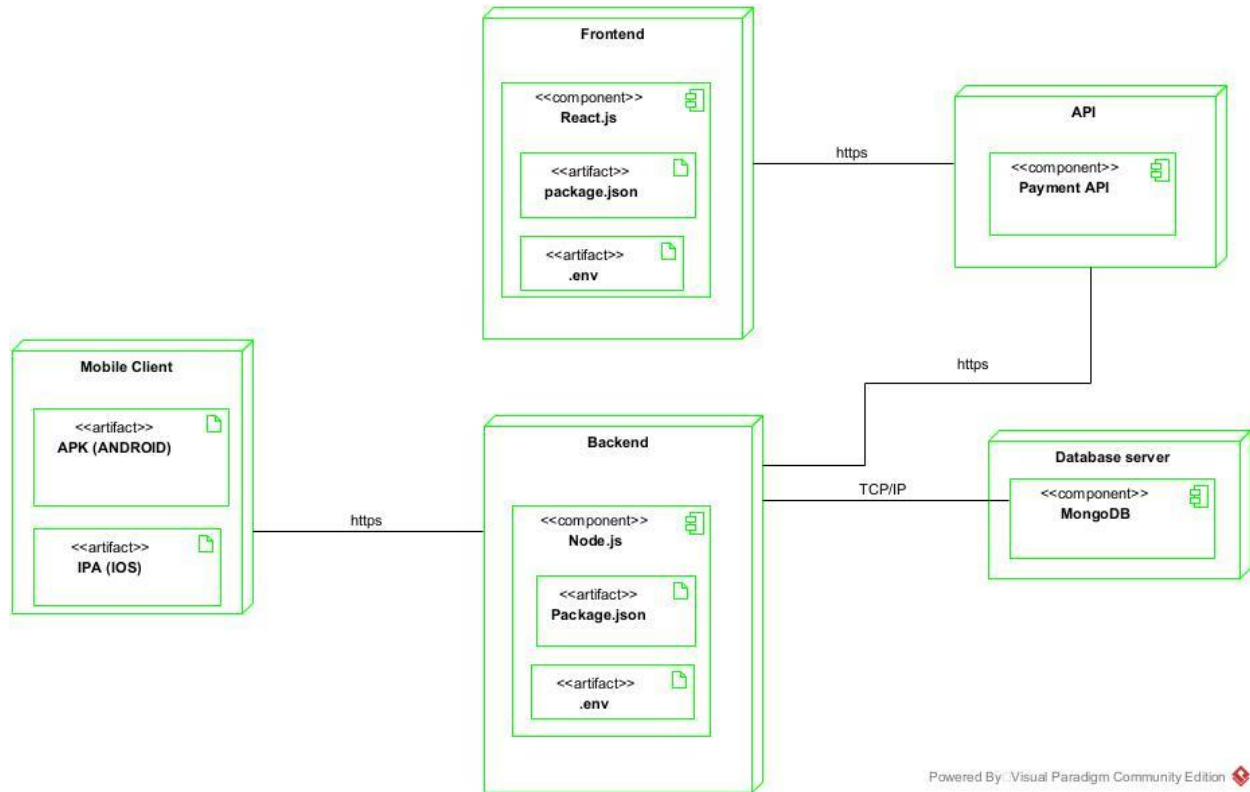


Figure 23: CamEdu deployment diagram

2. COMPONENT DIAGRAM

a. Definition

Component diagrams are used to model the physical aspect of a system. Now the question is what are this physical aspect? They are elements such as Executables, libraries, files, document etc. which resides in a node. The component diagram does not describe the functionality of the system, but it describes the components used to make those functionalities.

b. Formalism

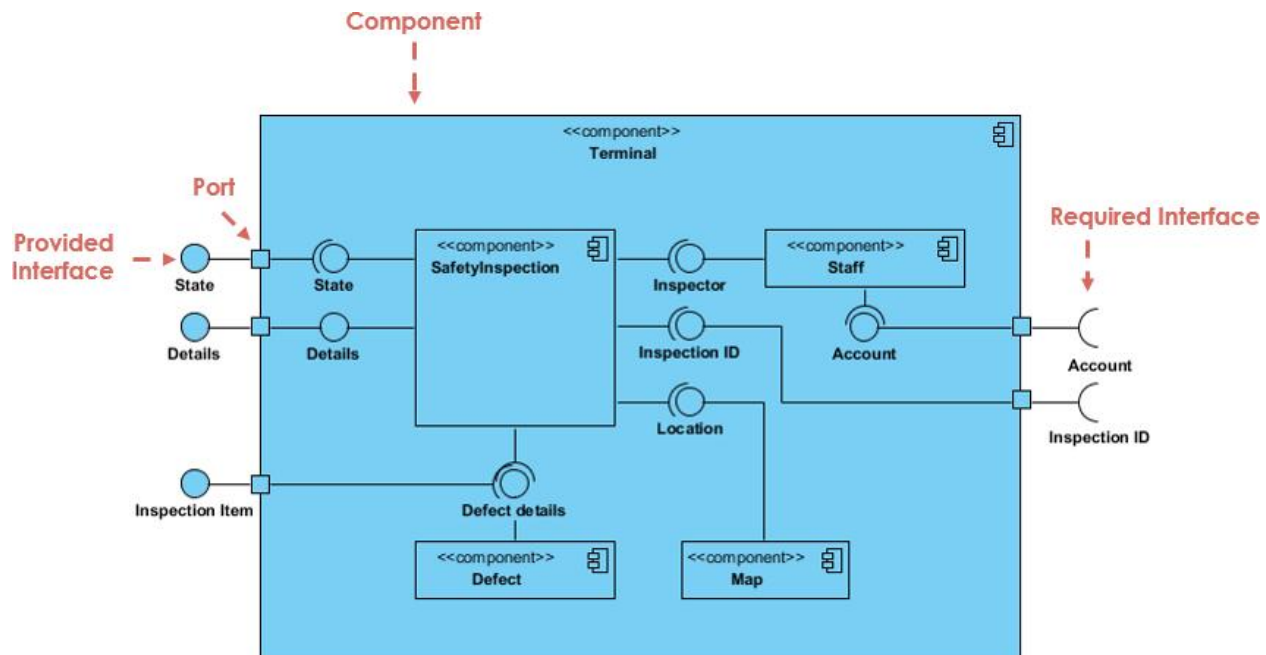
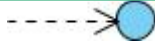
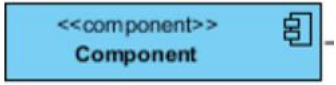


Figure 24: Component diagram formalism

c. Component of Component diagram

Table 23: Component of deployment diagram

Element	Notation	Description
Dependency		A dependency is a relationship that signifies that a single or a set of model elements requires other model elements for their specification or implementation.
Port		A port is often used to help expose required and provided interfaces of a component.
Interface		It is a point of interaction between a component and its environment or other components. It specifies the services that a component provides or requires from other components
Component		It represents a self-contained and replaceable unit that interacts with other components via provided and required interfaces



d. CamEdu Component diagram

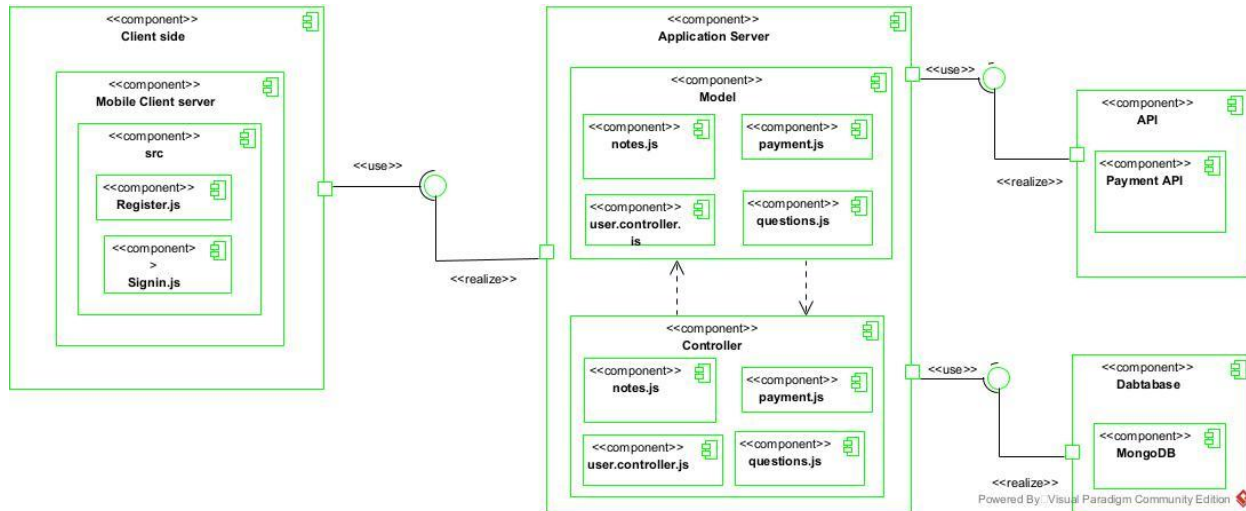
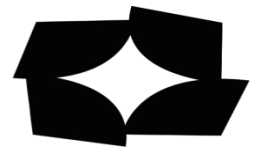


Figure 25: CamEdu component diagram



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CONCLUSION

During our realization phase, we successfully developed and implemented our application. To achieve this, we utilized our analysis and conception phase. We also created deployment and component diagrams, which illustrate the modules, files, and assets, as well as how different elements interact with each other within our system. Our next step is to move onto the functionalities testing phase, where we will assess the efficacy of the various modules present in our app for different users.



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DOCUMENT VI: TEST OF FUNCTIONALITIES



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



PREAMBLE

In this report, we will explore the different aspects of the test of functionalities and its significance in the software development life cycle.

CONTENT

INTRODUCTION

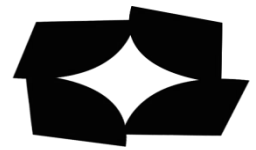
I. TEST OF FUNCTIONALITIES

II. TEST SHOWCASES

CONCLUSION

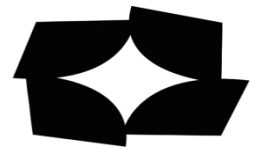


CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



INTRODUCTION

The test of functionalities plays a crucial role in ensuring the quality and reliability of software systems. By thoroughly examining each functionality and validating its behavior, developers can identify potential issues early in the development process. This helps in reducing the overall cost and effort required for bug fixing and maintenance.



I. TEST OF FUNCTIONALITIES

1. What is Unit Testing

Unit testing involves testing the smallest parts of an application, such as functions or methods, independently from the rest of the application. The primary goal is to validate that each unit of the software performs as expected.

2. Unit Testing Importance

Unit testing ensures that each component of the application works independently and correctly. It helps catch bugs early in the development process and guarantees that future changes don't break existing functionality.

3. Testing Strategy

We implemented **unit testing** to isolate and test individual parts of the app, specifically the API endpoints and backend services. Unit testing provides fast feedback on code quality and behavior.

4. Tools Used

- **Jest:** Testing framework used for running unit tests.
- **Supertest:** For testing API requests.
- **Mockingoose:** Used in some instances to mock MongoDB models for tests.

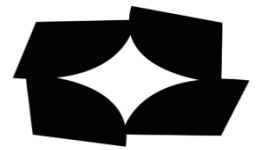
5. Key Functionalities Tested

a. User Registration

- Tests ensure that users can successfully register with valid details.
- Handles test cases where invalid inputs or missing fields cause registration failure.

b. User Authentication

- Tests confirm that users can log in with correct credentials.
- Includes scenarios where login fails due to invalid credentials (e.g., wrong password).



II. TEST SHOW CASES

a. Passing Unit Tests

- User registration.

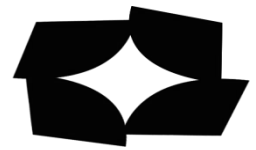
```
✓ should register a new user successfully (188ms)
✓ should return 400 for missing required fields
✓ should return 409 for duplicate email
✓ should register teacher with all required fields (129ms)
```

- User Authentication: Shows successful login and token generation when valid credentials are provided

```
✓ should login user successfully (134ms)
{
  _id: new ObjectId('68dae94804bf26094eb52e4e'),
  name: 'Test User',
  email: 'test@example.com',
  password: '$2b$10$0yGgOU4wsdtFSUTkxz30y.Ot1FQgp/TsfL4uhlA7Pe68WNz0Rs8i.',
  role: 'student',
  level: [],
  qualification: [],
  createdAt: 2025-09-29T20:17:12.949Z,
  updatedAt: 2025-09-29T20:17:12.949Z,
  __v: 0
}
user data wrongpassword password
✓ should return 400 for invalid credentials (147ms)
✓ should return 404 for non-existent user
✓ should return 400 for missing email or password
GET /api/users/get-users
```



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



CONCLUSION

Unit testing played a vital role in ensuring that core functionalities, such as user registration, login, and booking, work as expected. By focusing on isolated testing, we ensured that the app components were reliable and bug-free. In the final phase, we will provide an installation and user guide that will explain how to install and use CamEdu.



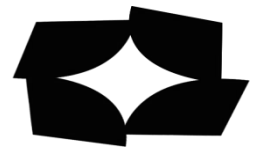
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DOCUMENT VII: INSTALLATION GUIDE
AND USER GUIDE



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PREAMBLE

The aim of the user manual is to furnish our platform users with systematic guidance on installing and utilising the system.

CONTENT

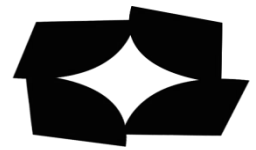
INTRODUCTION

- I. INSTALLATION OF THE APPLICATION
- II. SHOWCASE

CONCLUSION

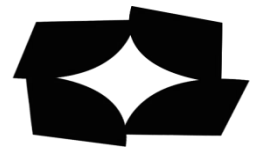


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INTRODUCTION

This is the ultimate stage of our report. In this phase, we will dissect the requirements for our system, the essential installation process, system and feature accessibility in a step-by-step approach to aid first-time users with easy platform setup. Each process will be accompanied by relevant images. Subsequently, we shall present our application by showcasing the various screens.



I. INSTALLATION OF THE APPLICATION

1. System requirements

1. For Developers:

- **Operating System:** Windows, macOS, or Linux
- **Node.js:** Version 14.x or later
- **npm or yarn:** Latest version (for package management)
- **Expo CLI:** Latest version (for running the app)
- **MongoDB:** Latest version (for database)
- **Visual Studio Code** or any other preferred text editor/IDE
- **Git** (for cloning the repository)

2. For Users:

- **Device:** Android phone
- **Operating System:** Requires Android 8.0 or later.
- **Internet Connection**

2. Installation Guide (For Developers)

2.1. Backend Installation

Step 1: Clone the repository

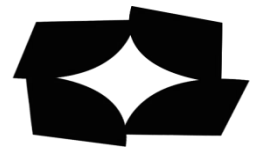
```
git clone https://github.com/Wansi237/CamEduBackend.git  
cd CamEduBackend
```

Step 2: Install dependencies

```
npm install
```



CONCEPTION AND REALISATION OF AN INTEGRATED NATIONAL BASED LEARNING APPLICATION FOR CAMEROON



Step 3: Set up environment variables

- Create a .env file in the root of your backend project folder and add the following:

```
MONGODB_CONNECTION_STRING=mongodb://localhost:27017/CamEdu
JWT_SECRET_KEY=your_jwt_secret_key

CLOUDINARY_CLOUD_NAME=your_cloudinary_cloud_name
CLOUDINARY_API_KEY=your_cloudinary_api_key
CLOUDINARY_CLOUD_SECRET=your_cloudinary_api_key
```

- Replace with your variables

Step 4: Run the server

```
npm run dev
```

2.2. Frontend Installation (React Native)

Step 1: Clone the repository

```
git clone https://github.com/Wansi237/CamEduFrontend.git
cd CamEduFrontend
```

Step 2: Install dependencies

```
npm install
```



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- Replace with yours

Step 4: Run the expo project with Expo

```
npm expo start
```

II. SHOWCASE

1. Registration:

- Open the app and click on Get Started button
- Fill in your name, email, address, password and confirm password
- Click submit to Create Account

Figure 26: Registration screen

2. Login:

- Open the app and enter your registered email and password.
- Click Log in to access your account

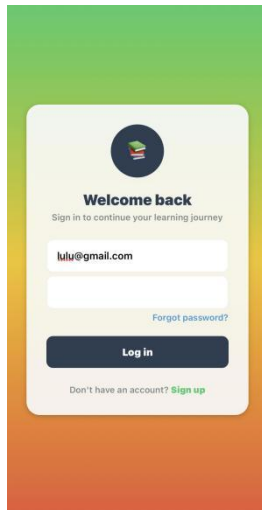


Figure 27: Login screen

3. Subjects view

Step 1: Select on Add Subject

Step 2: Click on the selected subjects

Step 3: View Overall progress and manage account

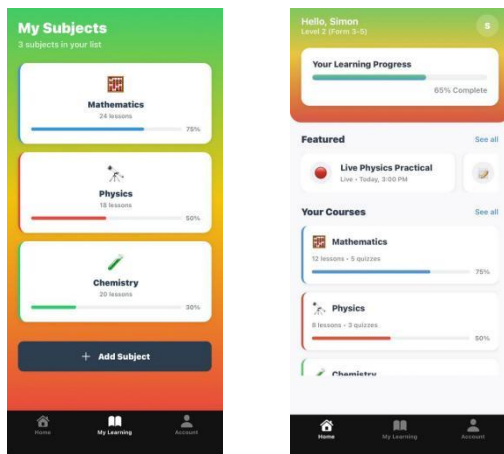
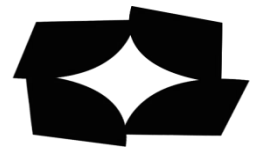


Figure 28: Subjects screens

Figure 47: Booking a transit step form 1



GENERAL CONCLUSION

As our internship at REALIZE comes to an end, we reflect on the various accomplishments we have achieved during this challenging yet enriching period. Managing challenges, learning new skills, and collaborating within a professional environment has provided us with invaluable experience and strengthened our teamwork abilities. For our project, we focused on the **“Conception and Realization of an Integrated National-Based Learning Application for Cameroon,”** named CamEdu. Our primary objective was to develop a platform that centralizes educational resources, facilitates access to learning materials, and tracks learner progress across Cameroon. We started with a detailed analysis of user needs and system requirements, followed by the application of the UML-2TUP methodology to design and structure the system. Using Visual Paradigm, we created comprehensive diagrams to guide the development process. To bring CamEdu to life, we employed **React Native** for the front-end, **Node.js** for the back-end, and **MongoDB** as our database. The system allows learners to access courses, track their progress, and participate in assessments, while educators and administrators can manage content, monitor learner performance, and generate reports. By providing a user-friendly, integrated, and scalable platform, CamEdu aims to enhance learning experiences and promote digital education nationwide. This project has the potential to transform the educational landscape in Cameroon by providing a reliable and accessible learning platform. Moving forward, our commitment to learning and improvement will continue, allowing us to enhance CamEdu further and expand its impact on learners and educators alike.



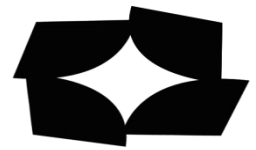
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ANNEXES



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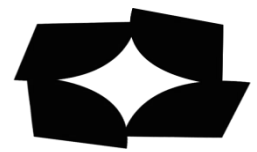
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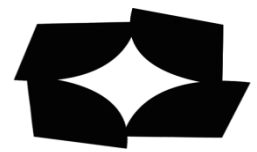
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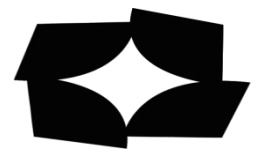


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